Cascaded $\chi^{(2)}{:}\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$ in a Doubly-Resonant Dielectric Metasurface

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1 Setting and goal

The objective is to engineer a metasurface whose effective third-order nonlinear susceptibility, *driven only at the fundamental frequency* ω_1, exceeds the bulk $\chi^{(3)}$ of the constituent material by orders of magnitude. The mechanism is the well-known *cascade* of two $\chi^{(2)}$ processes: $\omega_1+\omega_1 \to 2\omega_1$ followed by $2\omega_1 - \omega_1 \to \omega_1$. When the second-harmonic (SH) channel is *off-resonant* (or phase-mismatched), the SH amplitude follows the fundamental adiabatically and its back-action produces a Kerr-like response on the fundamental [1, 2, 3].

The metasurface platform we have in mind consists of an array of high-index dielectric resonators ($LiNbO_3$, GaAs, GaP, Si) supporting two co-localised resonances:

- a fundamental resonance at ω_1 – a *quasi*-bound state in the continuum (qBIC) with controlled radiative coupling $\gamma_1^{(\text{rad})}$ to free space, designed for in-coupling at ω_1;
- a second resonance at $\omega_2 \approx 2\omega_1$ – a symmetry-protected BIC (or strongly under-coupled qBIC) whose total decay γ_2 is dominated by intrinsic absorption, giving the largest accessible quality factor $Q_2 = \omega_2/\gamma_2$.

The relevant tensor of $\chi^{(2)}$ is patterned in space (via the geometry of the meta-atoms, by periodic poling, or by symmetry-breaking perturbations) so that the SHG overlap integral does not vanish (Section 8). High-Q metasurface SHG of exactly this kind has been demonstrated in AlGaAs and $LiNbO_3$ [16, 17, 18]. These experiments measure the SH radiation leaving the sample and therefore probe the $\chi^{(2)}$-driven up-conversion channel; the *cascaded* $\chi^{(3)}$ regime is the complementary observable – an intensity-dependent phase shift on the FM in conditions where the SH radiation is intentionally suppressed – and has not, to our knowledge, been isolated experimentally in a metasurface platform. A literature audit before submission is warranted, since $\chi^{(2)}{:}\chi^{(2)}$ cascading in waveguides and bulk PPLN [2, 3, 4] is well-established and the metasurface extension is a natural step.

2 Conventions

Several factors of 2 and 3 in the final answer come from book-keeping in the field/polarisation conventions; we fix them once and for all.

Real fields. Physical real fields are written as a sum of complex envelopes in Boyd's convention [5], without a prefactor $\frac{1}{2}$,

$$\mathcal{E}_i(\mathbf{r},t) = \sum_n \big[E_{n,i}(\mathbf{r},t)\, e^{-i\omega_n t} + \text{c.c.}\big], \qquad \mathcal{P}_i(\mathbf{r},t) = \sum_n \big[P_{n,i}(\mathbf{r},t)\, e^{-i\omega_n t} + \text{c.c.}\big], \tag{1}$$

with $E_{n,i}, P_{n,i}$ slowly varying compared with $1/\omega_n$. The polarisation relations below, and the tabulated values $\chi^{(2)} = 2d$ and $\chi^{(3)}$ used later, belong to this convention. (An earlier version of this note wrote a prefactor $\frac{1}{2}$ here while using these relations; the mismatch does not affect $\chi^{(3)}_{\mathrm{casc}}$, which is a ratio of material constants, but it does affect the energy reading of the modal amplitudes, Section 3.)

Susceptibility tensors (Boyd convention). With this field convention,

$$P_i^{(2)}(\omega_\sigma) = \varepsilon_0 \sum_{(p,q)} \chi^{(2)}_{ijk}(\omega_\sigma; \omega_p, \omega_q)\, E_j(\omega_p)\, E_k(\omega_q), \tag{2}$$

where the sum runs over distinct ordered pairs of frequencies summing to ω_σ. Indices are summed by Einstein convention.

For the two processes at hand,

$$\boxed{P_i^{(2)}(2\omega_1) = \varepsilon_0\, \chi^{(2)}_{ijk}(2\omega_1; \omega_1, \omega_1)\, E_j(\omega_1)\, E_k(\omega_1)} \qquad \text{(SHG)} \tag{3}$$

$$\boxed{P_i^{(2)}(\omega_1) = 2\varepsilon_0\, \chi^{(2)}_{ijk}(\omega_1; 2\omega_1, -\omega_1)\, E_j(2\omega_1)\, E_k^*(\omega_1)} \qquad \text{(DFG)} \tag{4}$$

The factor 2 in DFG counts the orderings $(2\omega_1, -\omega_1)$ and $(-\omega_1, 2\omega_1)$. **This is the factor missing in the original note**, and dropping it violates Manley–Rowe (Section 5).

The native Kerr response on the fundamental, for comparison, is

$$P_i^{(3)}(\omega_1) = 3\varepsilon_0\, \chi^{(3)}_{ijkl}(\omega_1; \omega_1, \omega_1, -\omega_1)\, E_j(\omega_1) E_k(\omega_1) E_l^*(\omega_1), \tag{5}$$

with the factor 3 counting the distinct orderings of $(\omega_1, \omega_1, -\omega_1)$.

We assume Kleinman symmetry (full permutation symmetry across all index pairs), valid for both $LiNbO_3$ and GaAs at telecom-band fundamentals well below the electronic gap. In particular,

$$\chi^{(2)}_{ijk}(2\omega_1; \omega_1, \omega_1) = \chi^{(2)}_{kij}(\omega_1; 2\omega_1, -\omega_1) \equiv \chi^{(2)}_{ijk}(\mathbf{r}), \tag{6}$$

i.e. a single symmetric tensor $\chi^{(2)}_{ijk}(\mathbf{r})$ in what follows.

3 Modes and energy normalisation

Let $\mathbf{f}_n(\mathbf{r}), \mathbf{h}_n(\mathbf{r})$ be the eigenmodes of the linear, lossless, dispersive Maxwell problem,

$$\nabla \times \nabla \times \mathbf{f}_n(\mathbf{r}) = \frac{\omega_n^2}{c^2}\, \varepsilon_r(\mathbf{r}, \omega_n)\, \mathbf{f}_n(\mathbf{r}), \tag{7}$$

with $\mathbf{f}_n$ a complex vector field encoding the polarisation pattern of mode n (and the magnetic mode $\mathbf{h}_n = \nabla \times \mathbf{f}_n/(i\omega_n\mu_0) = -i\nabla \times \mathbf{f}_n/(\omega_n\mu_0)$, following Faraday's law $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ under the $e^{-i\omega t}$ convention of §2). Note that both the *vectorial* nature of $\mathbf{f}_n$ and the dispersion of ε_r are essential; the original note's scalar, dispersionless $f(\mathbf{r})$ drops information needed for the tensorial overlap below.

The mode is energy-normalised so that

$$\boxed{\int \left[\, \varepsilon_0\, \widetilde{\varepsilon}_n(\mathbf{r})\, |\mathbf{f}_n|^2 + \mu_0 |\mathbf{h}_n|^2 \,\right] d^3r = 1, \qquad \widetilde{\varepsilon}_n(\mathbf{r}) \equiv \left.\frac{\partial(\omega\, \varepsilon_r(\mathbf{r}, \omega))}{\partial \omega}\right|_{\omega=\omega_n}} \tag{8}$$

(Brillouin's expression for the energy density of a time-harmonic field in a dispersive medium [6, §80]; in Boyd's convention the time average of $\mathcal{E}^2$ is $2|E|^2$, so the energy density is $\varepsilon_0\widetilde{\varepsilon}|E|^2+\mu_0|H|^2$ with no factor $\frac{1}{2}$). Defining the modal amplitude $a_n(t)$ by

$$E_{n,i}(\mathbf{r},t) = a_n(t)\, f_{n,i}(\mathbf{r}), \tag{9}$$

the time-averaged energy stored in mode n is exactly $|a_n|^2$ (so a_n has units of $\sqrt{\text{J}}$ and $|a_n|^2$ has units of J).

Projection norm. The quantity that appears when Maxwell's equations are projected onto a mode (Section 5) is

$$N_n \;\equiv\; \tfrac{1}{2}\int\!\big[\,\varepsilon_0\,\widetilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2\,\big]\,d^3r, \tag{10}$$

which reduces to $\int \varepsilon_0\varepsilon_r|\mathbf{f}_n|^2\,d^3r$ for a non-dispersive medium (the electric and magnetic energies of an eigenmode are equal) and equals $\frac{1}{2}$ in the normalisation (8). In general the stored energy is $U_n = 2N_n|a_n|^2$. We keep N_n explicit in the principal results so that they hold in any normalisation.

For non-dispersive lossless dielectrics one may replace $\widetilde{\varepsilon}_n \to \varepsilon_r(\mathbf{r},\omega_n) \equiv n^2(\mathbf{r},\omega_n)$. **Do not use a single n for both modes**: in $LiNbO_3$ at $\lambda_1 = 1550$ nm, $n_o(\omega_1) \simeq 2.21$ and $n_o(2\omega_1) \simeq 2.26$ (a $\sim 2\%$ difference), and the magnitude of $\chi^{(2)}_{ijk}$ also differs at the two frequencies via Miller's rule.

4 Polarisation-tensor derivation: $\chi^{(3)}_{\text{casc}}$ from $\chi^{(2)}\cdot G\cdot\chi^{(2)}$

Before introducing modal amplitudes it is instructive to derive $\chi^{(3)}_{\text{casc}}$ directly from the polarisation expansion and Maxwell's equations, without invoking coupled-mode theory. This route makes the structural form $\chi^{(3)}_{\text{casc}} \sim \chi^{(2)}\cdot G(2\omega_1)\cdot\chi^{(2)}$ manifest, exposes a non-local kernel that single-mode TCMT silently collapses, and shows that the temporal coupled-mode result of Section 5 is the same calculation re-expressed in a single-resonance approximation.

The SH field as the response to a $\chi^{(2)}$ source. At leading order in the nonlinearity the SH field is determined by Maxwell's equations with the SHG polarisation (3) as a source,

$$\big[\nabla\times\nabla\times\; - k_2^2\,\varepsilon_r(\mathbf{r},2\omega_1)\big]\mathbf{E}(2\omega_1) = k_2^2\,\mu_0 c^2\,\mathbf{P}^{\text{NL}}(2\omega_1), \qquad k_2 \equiv 2\omega_1/c. \tag{11}$$

Define $\overline{\overline{G}}(\mathbf{r},\mathbf{r}';2\omega_1)$ as the linear, retarded, dispersive dyadic Green's function for the operator on the left – i.e. the field at $\mathbf{r}$ produced by a unit polarisation source at $\mathbf{r}'$ in the same structure. Substituting (3),

$$\boxed{E_j(\mathbf{r},2\omega_1) = k_2^2\int d^3r'\; G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1)\,\chi^{(2)}_{lmn}(\mathbf{r}')\,E_m(\mathbf{r}',\omega_1)\,E_n(\mathbf{r}',\omega_1).} \tag{12}$$

This is the *only* step that goes beyond the susceptibility tensors: it tells us how the SH field is geometrically related to its $\chi^{(2)}$ source in the chosen structure (bulk medium, waveguide, metasurface, photonic crystal, ...).

Cascaded polarisation at ω_1. Substituting (12) into the DFG term (4) gives, with summation over repeated indices,

$$\boxed{P_i^{\mathrm{casc}}(\mathbf{r},\omega_1) = 2\varepsilon_0\, k_2^2 \int d^3r'\; \underbrace{\chi^{(2)}_{ijk}(\mathbf{r})\, G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1)\, \chi^{(2)}_{lmn}(\mathbf{r}')}_{\equiv\, \Xi^{\mathrm{casc}}_{ikmn}(\mathbf{r},\mathbf{r}')}\, E_m(\mathbf{r}',\omega_1)\, E_n(\mathbf{r}',\omega_1)\, E_k^*(\mathbf{r},\omega_1).} \tag{13}$$

Two structural features of (13) are worth emphasising before specialising.

The cascade is non-local. In general $P_i^{\mathrm{casc}}(\mathbf{r},\omega_1)$ depends on the FM field at a different point $\mathbf{r}'$ through the SH Green's function. A genuinely *local* effective $\chi^{(3)}(\mathbf{r})$ exists only when $G(\mathbf{r},\mathbf{r}';2\omega_1)$ is sufficiently sharply peaked at $\mathbf{r}=\mathbf{r}'$, or when the FM field varies slowly on the support of G. This is automatic in bulk media (translational invariance shrinks the kernel to a Δk-dependent local response) and in single-mode metasurface unit cells, but *not* in regimes where, e.g., the FM is a tightly focused beam probing a weakly coupled lattice resonance. In such regimes the local-$\chi^{(3)}$ description breaks down even though the underlying cascaded mechanism is unchanged.

The factor 2/3 from the Boyd degeneracies is universal. Comparing (13) with the native-Kerr definition $P^{(3)}(\omega_1) = 3\varepsilon_0\chi^{(3)}|E|^2E$ shows that the cascaded $\chi^{(3)}$ inherits a factor 2/3 relative to a naive product $|\chi^{(2)}|^2/G$, simply because the DFG channel has degeneracy 2 while the SPM channel has degeneracy 3. This shifts the answer in *both* the bulk and resonant limits below.

Two specialisations. Different physical contexts correspond to different forms of G:

- **(a) Bulk, phase-mismatched.** A translationally invariant medium has $G(\mathbf{r},\mathbf{r}';2\omega_1)$ that depends only on $\mathbf{r}-\mathbf{r}'$; in $\mathbf{k}$-space it has a smooth dependence peaked at $|\mathbf{k}| = 2\omega_1 n_2/c$. For plane-wave fundamental fields with wave-vector mismatch $\Delta\mathbf{k} = \mathbf{k}_2 - 2\mathbf{k}_1 \neq 0$ the integral in (13) collapses to a local response. The SVEA reduction [2, 3, 5] proceeds as follows. Writing the fields as $E(\omega_n,z) = A_n(z)\, e^{ik_n z}$ with slowly varying envelopes A_n, the wave equation (11) at $2\omega_1$ with the SHG polarisation (3) as source reduces (after dropping $\partial_z^2 A_2$ relative to $k_2\partial_z A_2$) to [5, §2.4]

$$\frac{dA_2}{dz} \;=\; \frac{i\,\omega_2}{2\,n_2\,c}\,\chi^{(2)}\,A_1^2\,e^{-i\Delta k z}. \tag{14}$$

  For undepleted pump (A_1 =const) and $A_2(0)=0$ this integrates to

$$A_2(z) \;=\; \frac{\omega_2\,\chi^{(2)}\,A_1^2}{2\,n_2\,c\,\Delta k}\left(1-e^{-i\Delta k z}\right). \tag{15}$$

  Substituting (15) into the DFG polarisation (4) and projecting the result back into the SVEA equation at ω_1, $dA_1/dz = (i\omega_1/(2n_1c))\cdot P^{\mathrm{NL}}(\omega_1)/\varepsilon_0\cdot e^{-ik_1 z}$, gives an intensity-dependent term $dA_1/dz \supset iK_{\mathrm{casc}}|A_1|^2A_1$ with

$$K_{\mathrm{casc}} \;=\; \frac{\omega_1\,\omega_2}{2\,n_1\,n_2\,c^2\,\Delta k}\,|\chi^{(2)}|^2 \;\times\; (\text{phase factor}).$$

  Identifying with the effective-Kerr SVEA equation $dA_1/dz = i\,(3\omega_1/(2n_1c))\,\chi^{(3)}_{\mathrm{casc}}\,|A_1|^2A_1$ (the factor 3 is the SPM degeneracy of Section 2) gives $\chi^{(3)}_{\mathrm{casc}} = K_{\mathrm{casc}}\,2n_1c/(3\omega_1)$; the explicit n_1 from the FM projection cancels against the n_1 in K_{casc}, leaving only the SH refractive index:

$$\boxed{\chi^{(3)}_{\mathrm{casc,\,bulk}}(\omega_1) \;=\; -\,\frac{2\,\omega_1}{3\,c\,n_2\,\Delta k}\,|\chi^{(2)}|^2\,\mathcal{F}(\Delta k\,L),} \tag{16}$$

with $\mathcal{F}(\Delta kL)$ a dimensionless phase factor that averages to $\mathcal{F}\to 1$ for $\Delta kL \gg 1$ (broadband bulk cascading; the regime exploited in soliton compression) and to $\mathcal{F}\to \Delta kL/2$ for $\Delta kL \ll 1$ (where SHG conversion competes with the cascade). Dimensionally $[\omega_1/(c\,\Delta k)]$ is dimensionless and $[|\chi^{(2)}|^2/n_2] = \mathrm{m}^2/\mathrm{V}^2$ as required; the explicit factor 2/3 is the DFG-to-SPM Boyd degeneracy ratio inherited from (13). Note that only n_2 (the SH-mode index) survives in the end, paralleling the metasurface result (35) below. The cascaded contribution is sign-tunable through $\mathrm{sign}(\Delta k)$ and bounded by the coherence length $\pi/\Delta k$ over which the cascade builds up coherently before reversing. Equation (16) is used in Section 10 to estimate the bulk PPLN cascaded $\chi^{(3)}$ for $LiNbO_3$.

- **(b) Single resonance (the metasurface case).** Near an isolated SH quasi-normal mode at $\omega_2 \approx 2\omega_1$ with vectorial profile $\mathbf{f}_2(\mathbf{r})$ and decay rate γ_2, the QNM expansion [10, 12, 13] gives, near the pole,

$$G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1) \;=\; -\frac{\varepsilon_0\, c^2}{2\omega_2 N_2}\,\frac{f_{2,j}(\mathbf{r})\, f^*_{2,l}(\mathbf{r}')}{\Delta\omega + i\gamma_2/2} \;+\; (\text{background}), \tag{17}$$

with $\Delta\omega = 2\omega_1 - \omega_2$, a non-resonant background, and N_2 the projection norm (10). The residue follows from the mode expansion of the operator in (11), $G = \sum_n \varepsilon_0 c^2\, \mathbf{f}_n \mathbf{f}_n^*/[(\omega_n^2 - \Omega^2)N_n]$ at $\Omega = 2\omega_1$, with $\omega_2^2 - \Omega^2 \simeq -2\Omega\Delta\omega$ near the pole; for leaky modes $\mathbf{f}_2\mathbf{f}_2^*/N_2$ is replaced by the QNM form of [12, Eq. (31)], which coincides with it in the high-Q limit. It fixes the prefactor of (20) independently of the TCMT route of Sections 5–6, and the two agree. If the FM field is also dominated by a single mode, $E_m(\mathbf{r},\omega_1) = a_1\, f_{1,m}(\mathbf{r})$, substituting (17) into the kernel $\Xi^{\mathrm{casc}}_{ikmn}(\mathbf{r},\mathbf{r}') = \chi^{(2)}_{ijk}(\mathbf{r})\, G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1)\, \chi^{(2)}_{lmn}(\mathbf{r}')$ of (13) and using $E_m(\mathbf{r}',\omega_1)E_n(\mathbf{r}',\omega_1) = a_1^2 f_{1,m}(\mathbf{r}')f_{1,n}(\mathbf{r}')$ isolates two overlap integrals

$$M \equiv \int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\; d^3r, \qquad \overline{M} \equiv \int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{1,i}\, f_{2,j}\, f^*_{1,k}\; d^3r, \tag{18}$$

related by Kleinman symmetry as $\overline{M} = M^*$ (the integrand of $\overline{M}$ is the complex conjugate of the integrand of M under the symmetry $\chi^{(2)}_{ijk} = \chi^{(2)}_{kij}$ of Section 2). The integration over $\mathbf{r}'$ contracts $f^*_{2,l}(\mathbf{r}')$ with $\chi^{(2)}_{lmn}(\mathbf{r}')\, f_{1,m}(\mathbf{r}')\, f_{1,n}(\mathbf{r}')$ – precisely M – while the remaining factor at $\mathbf{r}$, $\chi^{(2)}_{ijk}(\mathbf{r})\, f_{2,j}(\mathbf{r})\, E^*_k(\mathbf{r}) = a_1^*\, \chi^{(2)}_{ijk}\, f_{2,j}\, f^*_{1,k}$, has the modal polarisation pattern of $\mathbf{f}_1(\mathbf{r})$ when projected appropriately. The cascaded polarisation is therefore

$$P^{\mathrm{casc}}_i(\mathbf{r},\omega_1) \;=\; -\frac{2\varepsilon_0^2\,\omega_1}{N_2}\,\frac{M\,|a_1|^2\, a_1}{\Delta\omega + i\gamma_2/2}\,\chi^{(2)}_{ijk}(\mathbf{r})\, f_{2,j}(\mathbf{r})\, f^*_{1,k}(\mathbf{r}), \tag{19}$$

Kerr-like in the modal sense: it carries the SPM time-dependence $|a_1|^2 a_1$ and inherits the single-pole Lorentzian $1/(\Delta\omega + i\gamma_2/2)$ from G. Projecting (19) onto $f^*_{1,i}(\mathbf{r})$ replaces the spatial factor by $\overline{M} = M^*$, so $\int f^*_{1,i} P^{\mathrm{casc}}_i\, d^3r = -2\varepsilon_0^2\omega_1 |M|^2\, |a_1|^2\, a_1/[N_2(\Delta\omega + i\gamma_2/2)]$; equating this to the modal projection of the native Kerr term $\int f^*_{1,i} P^{\mathrm{nat}}_i\, d^3r = 3\varepsilon_0\, \chi^{(3)}_{\mathrm{casc}}\, |a_1|^2\, a_1 \int |\mathbf{f}_1|^4\, d^3r$ identifies the effective $\chi^{(3)}$,

$$\boxed{\chi^{(3)}_{\mathrm{casc}} \;=\; -\frac{2\,\varepsilon_0\,\omega_1}{3}\cdot\frac{|M|^2}{N_2\,(\Delta\omega + i\gamma_2/2)\int |\mathbf{f}_1|^4\, d^3r}} \tag{20}$$

whose real part recovers (33) exactly in the adiabatic limit $|\Delta\omega| \gg \gamma_2$, and whose imaginary part is the cascaded two-photon-like loss derived in Section 6. The result is invariant under rescaling of either mode ($|M|^2 \propto |\mathbf{f}_1|^4|\mathbf{f}_2|^2$ against $N_2 \int |\mathbf{f}_1|^4$), and its prefactor 2/3 is the DFG-to-SPM degeneracy ratio: the residue of (17) and the factor $\frac{1}{2}$ of the slowly-varying projection cancel. The TCMT route of Sections 5–6 gives the same answer from the modal projection of Maxwell's equations.

Equivalence with the TCMT derivation. Using the resonant Green's function (17) directly in (12) gives $\mathbf{E}(\mathbf{r}, 2\omega_1) = a_2 \mathbf{f}_2(\mathbf{r})$ with

$$a_2 \;=\; -\,\frac{\varepsilon_0\,\omega_1 M}{N_2}\,\frac{a_1^2}{\Delta\omega + i\gamma_2/2}\,, \tag{21}$$

which is precisely the steady state of the TCMT equation (22) with the coupling $\beta = \varepsilon_0\omega_1 M/N_2$ of (27), derived there independently. **The temporal coupled-mode theory is therefore the same calculation as (13) written in mode-amplitude language**, with $G(2\omega_1)$ replaced by its single-pole Lorentzian.

What this reframing buys.

- The structural formula $\chi^{(3)}_{\text{casc}} \sim \chi^{(2)} \cdot G(2\omega_1) \cdot \chi^{(2)}$ makes clear that one can engineer the cascade through *any* feature of G at $2\omega_1$: phase mismatch, BIC, photonic-crystal band edge, polariton, slow light. The high-Q SH mode of Section 5 is one (very effective) choice, not the only one.
- The non-local kernel $\Xi^{\text{casc}}(\mathbf{r}, \mathbf{r}')$ becomes important whenever the FM field varies on the scale of the SH mode – relevant for lattice resonances, surface-bound modes, or metasurfaces probed by tightly focused beams. In those regimes the local single-mode picture used in Sections 5–8 should be replaced by a full evaluation of (13).
- The factor 2/3 from Boyd degeneracies is the same in bulk and resonant cases – a useful sanity check whenever comparing to a Stegeman-style bulk estimate [3].

5 Coupled-mode equations

Substituting the modal expansion into Maxwell's equations and projecting onto $\mathbf{f}_n^*$ (slowly-varying-envelope approximation; e.g. [7, 8]) gives, after adding phenomenological loss rates and the input port for the FM:

$$\boxed{\frac{da_2}{dt} = \big(i\Delta\omega - \tfrac{\gamma_2}{2}\big)a_2 + i\beta\, a_1^2} \tag{22}$$

$$\boxed{\frac{da_1}{dt} = -\tfrac{\gamma_1}{2}a_1 + i\beta^*\, a_1^*\, a_2 + i\alpha_3\,|a_1|^2 a_1 + \sqrt{\gamma_1^{(\text{rad})}}\, s_+} \tag{23}$$

where $\Delta\omega \equiv 2\omega_1 - \omega_2$, s_+ is the input wave amplitude normalised so that $|s_+|^2$ is incident power, and the input–output relation is $s_- = -s_+ + \sqrt{\gamma_1^{(\text{rad})}}\, a_1$. The total FM decay is $\gamma_1 = \gamma_1^{(\text{rad})} + \gamma_1^{(\text{nr})}$; for the SH mode we have, by design, $\gamma_2 \simeq \gamma_2^{(\text{nr})}$.

The SHG and DFG couplings carry the same coupling constant β. This is the Manley–Rowe / energy-conservation requirement and can be checked directly: in the lossless limit ($\gamma_n, s_+ \to 0$, no Kerr), differentiating $|a_n|^2$ from the bare TCMT equations (22)–(23) gives, per channel,

$$\frac{d|a_2|^2}{dt} = 2\,\mathrm{Im}\big[\beta\, a_1^{*2} a_2\big], \qquad \frac{d|a_1|^2}{dt} = -2\,\mathrm{Im}\big[\beta^{(1)}\, a_1^{*2} a_2\big], \tag{24}$$

where $\beta^{(1)}$ denotes the (a priori unknown) coupling appearing in the FM equation (23) (we wrote $i\beta^* a_1^* a_2$ there anticipating the result $\beta^{(1)} = \beta$). The total mode-energy rate is therefore

$$\frac{d}{dt}(|a_1|^2 + |a_2|^2) = 2\,\mathrm{Im}\big[(\beta - \beta^{(1)})\, a_1^{*2} a_2\big], \tag{25}$$

which vanishes for arbitrary a_1, a_2 if and only if $\beta^{(1)} = \beta$, identifying the SH-equation coupling and the FM-equation coupling as complex conjugates of a single β. The same conclusion follows from the Hamiltonian formulation in [8, Eqs. 8–9], where energy conservation is built in from the start. **The factor 2 in the DFG polarisation (4) is exactly what produces this equality** – without it the SHG and DFG couplings would differ by a factor of 2 and energy would not be conserved. The original note's "κ vs. $\kappa/2$" form violates Manley–Rowe.

Derivation of the modal coefficients β and α_3. The boxed equations (22)–(23) are obtained by projecting Maxwell's equations onto $\mathbf{f}_n^*$. Write the field in the band centred at Ω as $\mathcal{E} = a(t)\,\mathbf{f}_n(\mathbf{r})\,e^{-i\Omega t} + \text{c.c.}$ with a slowly varying, and the nonlinear polarisation in that band as $\mathbf{P}\,e^{-i\Omega t} + \text{c.c.}$. Keeping first-order time derivatives of the envelope (including the dispersive one, $\varepsilon_r(\Omega + i\partial_t) \simeq \varepsilon_r + i\varepsilon_r'\partial_t$), the positive-frequency part of $\nabla\times\nabla\times\mathcal{E} + c^{-2}\partial_t^2\mathcal{D}_{\text{lin}} = -\mu_0\partial_t^2\mathcal{P}$ projected onto $\mathbf{f}_n^*$ gives

$$\dot{a} \;=\; i(\Omega-\omega_n)\,a \;+\; \frac{i\,\Omega}{2N_n}\int \mathbf{f}_n^*\cdot\mathbf{P}\;d^3r, \tag{26}$$

with N_n the projection norm (10): the coefficient of $\dot{a}$ is $\frac{1}{2}\varepsilon_0(\varepsilon_r+\widetilde{\varepsilon})$, and $\int\varepsilon_0\varepsilon_r|\mathbf{f}_n|^2 = \int\mu_0|\mathbf{h}_n|^2$ for an eigenmode. Equation (26) is the standard first-order perturbation result [11]

$$\delta\omega = -\frac{\omega}{2}\,\frac{\int\mathbf{E}^*\cdot\delta\mathbf{P}\;d^3r}{\int\varepsilon_0\varepsilon_r|\mathbf{E}|^2\,d^3r},$$

applied to a nonlinear polarisation. Applied to the three sources of Section 2:

- SH band ($\Omega = 2\omega_1$, $\omega_n = \omega_2$, so $i(\Omega-\omega_n) = i\Delta\omega$), SHG source (3): $\int\mathbf{f}_2^*\cdot\mathbf{P} = \varepsilon_0 M a_1^2$, hence $\beta = \varepsilon_0\omega_1 M/N_2$.
- FM band, DFG source (4): $\int\mathbf{f}_1^*\cdot\mathbf{P} = 2\varepsilon_0 M^* a_1^* a_2$ (Kleinman), hence the term $i\beta_1^* a_1^* a_2$ with $\beta_1 = \varepsilon_0\omega_1 M/N_1$. The DFG degeneracy 2 cancels the $\frac{1}{2}$ of (26).
- FM band, SPM source $3\varepsilon_0\chi^{(3)}|E|^2E$: hence $\alpha_3 = (3\varepsilon_0\omega_1/2N_1)\int\chi^{(3)}_{ijkl}f^*_{1,i}f_{1,j}f_{1,k}f^*_{1,l}$.

In the normalisation (8), $N_1 = N_2 = \frac{1}{2}$ and

$$\boxed{\beta = \frac{\varepsilon_0\,\omega_1}{N_2}\,M = 2\,\varepsilon_0\,\omega_1\int\chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{2,i}(\mathbf{r})\,f_{1,j}(\mathbf{r})\,f_{1,k}(\mathbf{r})\;d^3r} \qquad [\text{units: J}^{-1/2}\text{s}^{-1}], \tag{27}$$

$$\boxed{\alpha_3 = \frac{3\,\varepsilon_0\,\omega_1}{2N_1}\int\chi^{(3)}_{ijkl}\,f^*_{1,i}\,f_{1,j}\,f_{1,k}\,f^*_{1,l}\;d^3r} \qquad [\text{units: J}^{-1}\text{s}^{-1}]. \tag{28}$$

($= 3\varepsilon_0\omega_1\int\chi^{(3)}_{ijkl}f^*_{1,i}f_{1,j}f_{1,k}f^*_{1,l}$ at $N_1 = \frac{1}{2}$). With $N_1 = N_2$ the SHG and DFG couplings are the same β, which is the Manley–Rowe statement (25); for unequal norms the conserved quantity is $N_1|a_1|^2 + N_2|a_2|^2$, i.e. half the stored energy $U_1 + U_2$. The same couplings follow from the conservative Hamiltonian $H = \beta^* a_1^{*2}a_2 + \beta\,a_1^2a_2^*$ with $i\dot{a}_n = \partial H/\partial a_n^*$. The factor 3 in α_3 is the SPM degeneracy from Section 2 and was absent in the original note.

Correction to an earlier version of this note. An earlier version normalised the modes with a prefactor $\frac{1}{2}$ in (8) ($N_n = 1$) and used $\beta = (\varepsilon_0\omega_1/4)M$ and $\alpha_3 = (3\varepsilon_0\omega_1/8)\int\chi^{(3)}|\mathbf{f}_1|^4$; evaluated at $N_n = 1$, (27)–(28) give $\varepsilon_0\omega_1 M$ and $(3\varepsilon_0\omega_1/2)\int\chi^{(3)}|\mathbf{f}_1|^4$, so both were $4\times$ too small. The step-by-step projection there had dropped a factor of 2 ($\beta = \varepsilon_0 k_2^2 c^2 M/4\omega_2$ instead of $/2\omega_2$) and then adopted a further factor $\frac{1}{2}$ as a convention. Since $\chi^{(3)}_{\text{casc}} \propto |\beta|^2/\alpha_3$, the boxed cascaded susceptibility and every enhancement ratio R derived from it were $4\times$ too small; the TCMT dynamics, being dimensionless, were unaffected. The corrected prefactor reproduces the exact bulk bound-wave cascade in the uniform-medium limit (Section 6) and has been checked numerically on inhomogeneous structures.

Equivalence with the polarisation-tensor derivation. The TCMT route here and the Green's-function route of Section 4 are the same calculation in different representations. Substituting the QNM Green's function (17) into (12) gives $\mathbf{E}(\mathbf{r}, 2\omega_1) = a_2\,\mathbf{f}_2(\mathbf{r})$ with the steady-state $a_2 = -\beta a_1^2/(\Delta\omega + i\gamma_2/2)$ of (22), with exactly the coupling (27): the residue of (17) and the projection (26) are the same mode expansion, so the cascaded $\chi^{(3)}$ of (20) and that of the adiabatic-elimination route, eq. (33), agree including the prefactor.

6 Adiabatic elimination and the effective $\chi^{(3)}$

Assume $|\Delta\omega| \gg \gamma_2/2$ and $|\dot{a}_2| \ll |\Delta\omega\, a_2|$ (slow modulation of the FM compared with the SH response time). Setting $da_2/dt \to 0$ in (22),

$$a_2 = \frac{i\beta\, a_1^2}{-i\Delta\omega + \gamma_2/2} = \frac{-\beta\, a_1^2}{\Delta\omega + i\gamma_2/2} \simeq \frac{-\beta\, a_1^2}{\Delta\omega}\left(1 - \frac{i\gamma_2}{2\Delta\omega}\right). \tag{29}$$

Substituting into (23),

$$\frac{da_1}{dt} = \Big[-\tfrac{\gamma_1}{2} - \underbrace{\tfrac{|\beta|^2\gamma_2}{2\Delta\omega^2}}_{\text{cascaded loss}}\,|a_1|^2\Big]a_1 + i\Big[\underbrace{\alpha_3 - \tfrac{|\beta|^2}{\Delta\omega}}_{\text{effective Kerr}}\Big]|a_1|^2 a_1 + \sqrt{\gamma_1^{(\mathrm{rad})}}\,s_+. \tag{30}$$

(i) Effective Kerr coefficient.

$$\boxed{\alpha_3^{\mathrm{eff}} = \alpha_3 - \frac{|\beta|^2}{\Delta\omega}} \tag{31}$$

The cascaded contribution can have *either* sign relative to the bare Kerr response, depending on whether $\Delta\omega \equiv 2\omega_1 - \omega_2$ is positive or negative. This sign-tunability is one of the hallmarks of cascaded nonlinearity: the same metasurface can be tuned through zero, used for self-defocusing, or used for self-focusing simply by walking the SH resonance through twice the FM frequency. This was used dramatically in soliton compression experiments [2, 9].

(ii) Two-photon-like loss. The cascaded loss coefficient – defined as the coefficient X in the bracket form $da_1/dt = [-\gamma_1/2 - X|a_1|^2]a_1 + \ldots$ on (13) above – is, in the adiabatic limit $|\Delta\omega| \gg \gamma_2/2$,

$$\gamma_3^{\mathrm{eff}} \;=\; \frac{|\beta|^2\,\gamma_2}{2\,\Delta\omega^2}, \tag{32}$$

with full Lorentzian form $\gamma_3^{\mathrm{eff}}(\Delta\omega) = |\beta|^2(\gamma_2/2)/[\Delta\omega^2 + (\gamma_2/2)^2]$ peaking at $\Delta\omega = 0$ with value $2|\beta|^2/\gamma_2$. It is the analogue of two-photon absorption (here governed by γ_2 rather than by $\mathrm{Im}\,\chi^{(3)}$) and is unavoidable as the price of the resonant enhancement. In the same convention, the cascaded loss-to-Kerr ratio at the design point is $\gamma_3^{\mathrm{eff}}/|\alpha_3^{\mathrm{eff}}| = \gamma_2/(2|\Delta\omega|)$.

To compare with bulk material parameters, invert the definition of α_3 in (28): a uniform bulk $\chi_{\mathrm{eff}}^{(3)}$ filling the mode produces $\alpha_3 = (3\varepsilon_0\omega_1/2N_1)\,\chi_{\mathrm{eff}}^{(3)}\int|\mathbf{f}_1|^4\,d^3r$. The cascaded coefficient is $-\beta_1^*\beta/\Delta\omega = -\varepsilon_0^2\omega_1^2|M|^2/(N_1N_2\Delta\omega)$ (this is $-|\beta|^2/\Delta\omega$ when $N_1 = N_2$), so the cascaded contribution corresponds to

$$\boxed{\chi_{\mathrm{casc}}^{(3)} = -\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2\,\Delta\omega}\,\frac{\left|\int\chi_{ijk}^{(2)}(\mathbf{r})\,f_{2,i}^*\,f_{1,j}\,f_{1,k}\,d^3r\right|^2}{\int|\mathbf{f}_1|^4\,d^3r}} \tag{33}$$

(N_1 cancels; $N_2 = \frac{1}{2}$ in the normalisation (8), where the prefactor reads $-4\varepsilon_0\omega_1/3\Delta\omega$). The result is invariant under rescaling of either mode.

Explicit n-dependence. The n that one might have expected in a "naïve" (plane-wave-style) derivation is hidden in the energy normalisation (8): at unit stored energy $|a_n|^2 = 1$ J, the typical field magnitude in the dielectric is

$$|\mathbf{f}_n(\mathbf{r})| \;\sim\; \frac{1}{\sqrt{2\,\varepsilon_0\, n_n^2\, V_{\text{eff}}}}, \tag{34}$$

where n_n is the effective refractive index seen by mode n (close to the bulk index at ω_n for a tightly confined dielectric mode) and V_{eff} is the mode volume. Writing $\mathbf{f}_n = \mathbf{u}_n/\sqrt{2\varepsilon_0 n_n^2 V_{\text{eff}}}$ with $\int |\mathbf{u}_n|^2 = V_{\text{eff}}$, and defining the dimensionless $\zeta_1 \equiv \int |\mathbf{u}_1|^4/V_{\text{eff}}$ and $\widetilde{\eta} \equiv |\int \hat{\chi}_{ijk} u^*_{2,i} u_{1,j} u_{1,k}|/V_{\text{eff}}$ ($\hat{\chi} = \chi^{(2)}/|\chi^{(2)}|$; perfect overlap $\widetilde{\eta} \to 1$), all four powers of n_1 cancel between numerator and denominator of (33) and one ε_0 cancels against its prefactor, giving the final explicit form

$$\boxed{\chi^{(3)}_{\text{casc}} \;\simeq\; -\frac{2\,\omega_1}{3\,n_2^2\,\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\text{geom}}\,, \qquad \eta_{\text{geom}} \equiv \frac{\varepsilon_0 n_2^2 |M|^2}{|\chi^{(2)}|^2 N_2 \int |\mathbf{f}_1|^4} = \frac{\widetilde{\eta}^{\,2}}{\zeta_1} \sim 0.01\text{–}0.5} \tag{35}$$

with η_{geom} a purely geometric overlap that does not depend on the mode normalisation. Three things are worth emphasising about this form:

- **Only n_2 (the SH-mode effective index) survives.** The n_1 powers cancel between numerator and denominator because $|\mathbf{f}_1|^4$ appears symmetrically in both, leaving the SH-mode normalisation $|\mathbf{f}_2|^2 \propto 1/(\varepsilon_0 n_2^2 V)$ as the only residue. Physically, modes in high-index material concentrate *less* field per unit stored energy, so the $\mathbf{f}_2$:$\mathbf{f}_1\mathbf{f}_1$ coupling is weaker by $1/n_2$.
- **The dielectric constant ε_0 does not appear** in the final result. This is a non-trivial consistency check: $\chi^{(3)}_{\text{eff}}$ is a material parameter (m^2/V^2 in SI), and dimensional analysis of $|\chi^{(2)}|^2$ alone gives the right units.
- **The prefactor 2/3 is not a convention.** It is the DFG-to-SPM degeneracy ratio, and it is fixed by the bulk limit below.

Uniform-medium limit. For a single plane-wave mode pair filling a periodic box ($\mathbf{f}_1 \propto e^{ik_1 z}$, $\mathbf{f}_2 \propto e^{2ik_1 z}$, so $\eta_{\text{geom}} = 1$), and a non-dispersive medium in which the SH mode at $2k_1$ has $\omega_2 = 2\omega_1 n_1/n_2$, (35) gives

$$\chi^{(3)}_{\text{casc}} = -\frac{2\omega_1|\chi^{(2)}|^2}{3n_2^2\Delta\omega} \;\simeq\; -\frac{2\,|\chi^{(2)}|^2}{3\,(n_2^2 - n_1^2)}, \tag{36}$$

which is the exact bound-wave cascade of a uniform medium: solving the Helmholtz equation for $E(2\omega_1)$ driven by (3) gives $E_2 = \chi^{(2)}E_1^2/(n_1^2 - n_2^2)$, and (4) then yields the same $\chi^{(3)}$. It also coincides with the bulk formula (16) at $\mathcal{F} = 1$, since $cn_2\Delta k \simeq \omega_1(n_2^2 - n_1^2)$. Thus η_{geom} is the fraction of the bulk bound-wave cascade that the resonator recovers, and $\eta_{\text{geom}} \le 1$ by the Cauchy–Schwarz inequality ($\widetilde{\eta}^{\,2} \le \zeta_1$). The resonator replaces the phase mismatch Δk by the frequency mismatch $\Delta\omega$; its advantage is that $|\Delta\omega|$ can be made far smaller than $c\Delta k/n_2$.

7 The bandwidth–enhancement sum rule

The peak enhancement is bounded above by the SH linewidth. The cleanest derivation is from the AC frequency response of the cascade. If $|a_1|^2$ is modulated at frequency Ω (so the FM intensity has a Fourier component $\delta|a_1|^2\,e^{-i\Omega t}$ on top of a DC value), the SH equation (22)

generates a corresponding SH-amplitude sideband $\delta a_2 = i\beta\,\delta(a_1^2)/(-i(\Delta\omega-\Omega)+\gamma_2/2)$, and the DFG back-action on the FM acts as an effective *frequency-dependent* Kerr coefficient

$$\alpha_3^{\text{eff}}(\Omega) \;=\; \frac{i\,|\beta|^2}{\gamma_2/2 + i(\Omega-\Delta\omega)}, \qquad |\alpha_3^{\text{eff}}(\Omega)| \;=\; \frac{|\beta|^2}{\sqrt{(\gamma_2/2)^2+(\Omega-\Delta\omega)^2}}. \tag{37}$$

The *Lorentzian* in Ω here is $|\alpha_3^{\text{eff}}(\Omega)|^2$, centred at $\Omega=\Delta\omega$ with FWHM $=\gamma_2$, peak $(2|\beta|^2/\gamma_2)^2 = 4|\beta|^4/\gamma_2^2$, and area

$$\int d\Omega\,|\alpha_3^{\text{eff}}(\Omega)|^2 \;=\; \frac{2\pi\,|\beta|^4}{\gamma_2}. \tag{38}$$

$|\alpha_3^{\text{eff}}(\Omega)|$ itself is the square root of this Lorentzian, so it is *broader*: $\text{FWHM}(|\alpha|) = \sqrt{3}\,\gamma_2$ (taking the square root widens by $\sqrt{3}$). Its peak is $|\alpha_3^{\text{eff}}|_{\text{peak}} = |\beta|^2/(\gamma_2/2) = 2|\beta|^2/\gamma_2$ at $\Omega=\Delta\omega$, and the peak-times-FWHM product

$$|\alpha_3^{\text{eff}}|_{\text{peak}} \times \text{FWHM}(|\alpha_3^{\text{eff}}|) \;=\; 2\sqrt{3}\,|\beta|^2, \tag{39}$$

is conserved as Q_2 varies (since $|\beta|^2 \propto |\chi^{(2)}|^2\eta_{\text{geom}}/n_2^2$ of Section 6 is set by material constants and overlap geometry alone, independent of Q_2). The cascaded coefficient has resonant gain when the FM modulation frequency matches the SH detuning.

In design terms, at the conventional operating point $|\Delta\omega|\sim\gamma_2$: $|\alpha_3^{\text{eff}}|_{\max}\sim|\beta|^2/\gamma_2$ and the usable bandwidth is $\Delta\Omega\sim\gamma_2$, so combining with (35) (using $\omega_1/\gamma_2 = Q_2/2$ since $\gamma_2=\omega_2/Q_2 = 2\omega_1/Q_2$):

$$\left|\chi^{(3)}_{\text{casc}}\right|_{\max} \;\sim\; \frac{2\,\omega_1}{3\,n_2^2\,\gamma_2}\,|\chi^{(2)}|^2\,\eta_{\text{geom}} \;=\; \frac{Q_2}{3\,n_2^2}\,|\chi^{(2)}|^2\,\eta_{\text{geom}}\,, \tag{40}$$

and (multiplying by $\Delta f_{\text{usable}}\sim\gamma_2/(2\pi)$, so that γ_2 cancels)

$$\boxed{\left|\chi^{(3)}_{\text{casc}}\right|_{\max} \times \Delta f_{\text{usable}} \;\lesssim\; \frac{\omega_1\,|\chi^{(2)}|^2\,\eta_{\text{geom}}}{3\pi\,n_2^2} \quad \text{(independent of } Q_2\text{).}} \tag{41}$$

Q_2 does not enter the product: **resonant enhancement redistributes nonlinear action into a tall, narrow peak; it does not increase the area under the curve.** This is the same rule that limits resonant linear absorption (oscillator-strength sum rule), purely classical, and it is what disqualifies high-Q double-resonance schemes from delivering "fast" $\chi^{(3)}$ in the ultrafast sense. For the nominal numbers below ($Q_2\sim 4\times10^3$ at $\lambda_1 = 1.55\,\mu$m, so $\gamma_2=\omega_2/Q_2 = 2\omega_1/Q_2$), $\Delta f_{\text{usable}}\sim\gamma_2/(2\pi)\simeq 97$ GHz: narrowband by Kerr-electronics standards (THz), broadband by photonic-cavity standards.

For non-resonant (phase-mismatched bulk) cascading the same sum rule applies with γ_2 replaced by the inverse coherence time $|\Delta k|\,v_g$, giving modest enhancement over a much larger bandwidth – the regime exploited in soliton compression and bulk all-optical switching.

Equation (37) describes the cascade response to a modulation of $|a_1|^2$ at frequency Ω (relevant for slow envelope modulation of the FM amplitude); the input is the pump intensity and the AC-peak/DC enhancement of $|\alpha_3^{\text{eff}}|$ is $|\Delta\omega|/(\gamma_2/2)$ asymptotically. A *distinct* transfer function (different input, different output) is treated in Section 12, item 6: there one modulates the SH-cavity detuning $\Delta\omega(t)$ at frequency Ω_{mod} – this is the cascade-based amplitude modulator – and its AC-peak/DC enhancement is $\sqrt{\Delta\omega_0^2+(\gamma_2/2)^2}/\gamma_2 \approx |\Delta\omega_0|/\gamma_2$ (eq. (66)), smaller by a factor of 2 than the FM-modulation result because the SH-detuning modulation only drives one sideband per Fourier component of $\Delta\omega(t)$.

8 Tensor structure and phase matching

The overlap that defines β,

$$\beta \propto \int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\, d^3r,$$

vanishes identically for several reasons, each of which must be addressed in the metasurface design:

1. **Bulk inversion symmetry.** If a meta-atom has inversion symmetry, $\chi^{(2)}_{ijk} \equiv 0$. Hence one needs a non-centrosymmetric crystal substrate ($LiNbO_3$, point group $3m$; GaAs/GaP, point group $\bar{4}3m$ (T_d); AlGaAs; ...) *and* a meta-atom geometry that does not impose inversion symmetry on the mode profile.

2. **Mode-symmetry mismatch.** Even with $\chi^{(2)}_{ijk} \neq 0$ in the material, the integrand has a definite parity under any spatial symmetry shared by both modes. If the FM mode is, say, magnetic-dipole-like (odd under a mirror) and the SH is electric-dipole-like (even), the integrand may be odd under that mirror and the integral vanishes. Quasi-BIC engineering is precisely the controlled breaking of one such symmetry to lift this constraint while keeping Q_2 large.

3. **Quasi-phase matching across the metasurface period.** For an extended metasurface, the integrand carries Bloch phase factors $e^{i(2\mathbf{k}_1-\mathbf{k}_2)\cdot\mathbf{R}}$. Periodic poling or spatial modulation of $\chi^{(2)}_{ijk}(\mathbf{r})$ across unit cells provides a reciprocal-lattice vector $\mathbf{G}$ that compensates the residual momentum $2\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{G} = 0$. In a single-unit-cell metasurface with both modes at $\mathbf{k}_\parallel = 0$ (Γ point), this momentum-matching is automatic and the only requirement is on the *internal* symmetry of the mode profiles.

For $LiNbO_3$ (point group $3m$), the non-zero independent components of $\chi^{(2)}_{ijk}$ in contracted notation are $d_{15} = d_{31}$, $d_{22} = -d_{21} = -d_{16}$, and d_{33}, with $|d_{33}| \simeq 25$ pm/V at 1064 nm being the largest. With the $\hat{z}$ axis parallel to the optic axis, the largest overlap is achieved by mode profiles with strong $|f_{1,z}|^2 f^*_{2,z}$ content over the meta-atom volume. Z-cut $LiNbO_3$ with field components engineered along the optic axis is therefore preferred. The same considerations apply, with the appropriate tensor structure, for $\bar{4}3m$ semiconductors (largest component $d_{14} = d_{25} = d_{36} \simeq$ 100 pm/V for GaAs).

For $LiNbO_3$, $\chi^{(2)} \equiv 2d_{33} \simeq 50$ pm/V $= 5 \times 10^{-11}$ m/V; for GaAs, $\chi^{(2)} \simeq 2 \times 10^{-10}$ m/V.

9 Metasurface implementation: BIC and quasi-BIC platforms

The high-Q_2 requirement is the key technical hurdle, and bound states in the continuum (BIC) provide a particularly elegant route [14]. A symmetry-protected BIC is a guided-wave-like solution that lies inside the radiation continuum but is *symmetry-forbidden* from coupling to plane waves at $\mathbf{k}_\parallel = 0$. Realistic structures are quasi-BICs (qBICs): finite arrays, slight disorder, or deliberate symmetry breaking convert the ideal BIC into a high-Q leaky resonance, whose Q scales as $1/\alpha^2$ with the asymmetry parameter α [15, 20]. Demonstrated Q factors in dielectric metasurfaces are routinely 10^3–10^4 and have reached $\sim 10^5$ in carefully optimised devices.

Two practical architectures:

(a) Two-mode qBIC pair. Both the FM and SH are qBICs of the same array, with the SH qBIC very close to the symmetry-protected limit (large Q_2) and the FM qBIC moderately broadened ($Q_1 \sim 10^2$–10^3) to allow efficient in-coupling at ω_1. The SH leakage to the

radiation continuum is irrelevant for the cascaded process – we *don't want* SH out-coupling here.

(b) FM Mie–qBIC + SH symmetry-protected BIC. The FM is a low-Q Mie resonance (dipolar, easy to pump); the SH lives in a true BIC mode of the array, with γ_2 set by absorption only. Best for maximum enhancement, hardest to fabricate.

In both cases the requirement on the SH mode is purely *internal*: high Q_2, large field overlap with the FM-squared, no need for any external connection.

A subtle point: the cascaded $\chi^{(3)}$ effect manifests at the FM, so the experimental signature is **intensity-dependent reflection/transmission at** ω_1 (self-phase modulation, intensity-dependent line shape, optical bistability for sufficient pump), not SH light leaving the sample. In fact, a *good* implementation produces *no measurable* SH radiation while the back-action on the fundamental is maximum. This is also the experimental fingerprint that distinguishes the cascaded process from the bare Kerr response.

10 A worked numerical estimate ($LiNbO_3$ metasurface)

Take

quantity	value
λ_1	$1.55\,\mu$m
ω_1	1.22×10^{15} rad/s
n_1, n_2	2.21, 2.26
n_2^2	$\simeq 5.1$
$\lvert\chi^{(2)}\rvert$ $(= 2d_{33})$	5×10^{-11} m/V
$\lvert\chi^{(3)}\rvert_{\text{native}}$	$\sim 2 \times 10^{-21}$ m^2/V^2
Q_2	4×10^3
$\gamma_2 = \omega_2/Q_2 = 2\omega_1/Q_2$	6.1×10^{11} rad/s ($\gamma_2/2\pi \simeq 97$ GHz)
$\Delta\omega$ (safely adiabatic, $\lvert\Delta\omega\rvert \gtrsim 2\gamma_2$)	$2\gamma_2 = 4\omega_1/Q_2 = 1.22 \times 10^{12}$ rad/s
η_{geom} (purely geometric overlap, eq. (35))	~ 0.3

Using the explicit form (35) with $\Delta\omega = 2\gamma_2$, so $\omega_1/\Delta\omega = Q_2/4$:

$$\begin{aligned} \left|\chi^{(3)}_{\text{casc}}\right| \;&\sim\; \frac{2\,\omega_1}{3\,n_2^2\,\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\text{geom}} \;=\; \frac{Q_2}{6\,n_2^2}\,|\chi^{(2)}|^2\,\eta_{\text{geom}} \\ &\sim\; \frac{4000}{6\times 5.1}\times 0.3\times|\chi^{(2)}|^2 \;\sim\; 39\times|\chi^{(2)}|^2, \end{aligned}$$

that is

$$\left|\chi^{(3)}_{\text{casc}}\right| \sim 39\times(5\times10^{-11})^2\ \text{m}^2/\text{V}^2 \;\simeq\; 9.8\times10^{-20}\ \text{m}^2/\text{V}^2,$$

i.e. roughly **49× the native bulk** $\chi^{(3)}$ of $LiNbO_3$, available over a usable bandwidth $\sim \gamma_2/(2\pi) \simeq 97$ GHz centred at the FM resonance (Section 7). The $1/n_2^2$ factor costs a factor of ~ 5 relative to a "naïve" estimate that drops it – it is not negligible, and an honest design has to compensate it with either larger η_{geom} or larger $Q_2/|\Delta\omega|$.

Operating-point sensitivity. Pushing $\Delta\omega$ from the safely adiabatic $2\gamma_2$ down toward the edge-of-validity $\Delta\omega \sim \gamma_2$ (Section 13, item 1) roughly doubles the enhancement to $R \sim 100$, at the cost of working at the edge of the adiabatic approximation where the cascaded TPA

(Section 6(ii)) becomes comparable to the Kerr response: indeed at $|\Delta\omega| = \gamma_2$, the loss-to-Kerr ratio $\gamma_3^{\text{eff}}/|\alpha_3^{\text{eff,casc}}| = \gamma_2/(2|\Delta\omega|) = 1/2$. We adopt the conservative $\Delta\omega = 2\gamma_2$, $R \sim 49$ throughout the rest of this note, and remind the reader of the factor-of-~ 2 headroom when interpreting numerical claims downstream.

What is and is not uncertain. The prefactor of (35) is exact: it is derived twice (Sections 4 and 6) and reproduces the bulk bound-wave cascade in the uniform limit. (Earlier versions of this note quoted $R \sim 12$ here, with an "$O(1)$ prefactor uncertainty"; that value was $4\times$ too small, see Section 5.) The remaining uncertainty is the assumed $\eta_{\text{geom}} \sim 0.3$, which is geometry-specific and is the single largest lever: R is linear in it. Settling it requires a full-wave evaluation of the modes and the overlap integral M in a specific meta-atom design.

With $Q_2 = 10^4$ (achievable in optimised qBIC $LiNbO_3$ metasurfaces) and the same $\eta_{\text{geom}} = 0.3$, the estimate gives $\sim 120\times$ enhancement at the cost of a ~ 40 GHz bandwidth.

For comparison, a phase-mismatched bulk PPLN slab at the same $|\chi^{(2)}|$ gives non-resonant cascaded $\chi^{(3)}$ via the SVEA-derived formula (16). Inserting $|\Delta k| \sim 3\times 10^5$ m^{-1} (corresponding to a $LiNbO_3$ coherence length $\pi/|\Delta k| \sim 10\,\mu$m at telecom; the bare $\Delta n \simeq 0.05$ between ω_1 and $2\omega_1$ gives $|\Delta k| \simeq 4\times 10^5$ m^{-1}, so this is a representative order-of-magnitude figure), $n_2 \simeq 2.26$, and $\omega_1/c \simeq 4\times 10^6$ m^{-1},

$$\left|\chi^{(3)}_{\text{casc, bulk}}\right| \;\sim\; \frac{2\cdot 4\times 10^6}{3\cdot 2.26\cdot 3\times 10^5}\cdot(5\times 10^{-11})^2 \;\sim\; 10^{-20}\ \text{m}^2/\text{V}^2,$$

roughly $5\times$ the native $\chi^{(3)}_{LiNbO_3}$, available over ~ 10 THz bandwidth – the regime exploited in soliton compression and bulk all-optical switching.

Comparing $|\chi^{(3)}|\times\Delta f$ products honestly: bulk PPLN gives $|\chi^{(3)}_{\text{casc, bulk}}|\times\Delta f \sim 10^{-20}\cdot 10^{13} = 10^{-7}$ m^{2}V^{-2}Hz; the metasurface at $Q_2 = 4\times 10^3$ gives $|\chi^{(3)}_{\text{casc, meta}}|_{\text{peak}}\times\Delta f \sim 9.8\times 10^{-20}\cdot 10^{11} \simeq 10^{-8}$ m^{2}V^{-2}Hz, about a factor ~ 10 *less* than bulk (the η_{geom} and $1/n_2^2$ penalties are not free; at the edge-of-adiabatic point $\Delta\omega = \gamma_2$ the gap narrows to ~ 5). The metasurface's appeal is therefore not in the integrated nonlinear strength but in **the chip-scale form factor and CW-low-threshold operation**: narrow-band high-peak operation in a ~ 100-nm-thick film, with μW-scale CW pumping rather than the > 100 W pulsed pumping of bulk PPLN cascading. Both regimes have their applications; the high-Q regime is suited to CW-pumped bistability, parametric oscillation thresholds and frequency-comb stabilisation, rather than to ultrafast switching.

11 Comparison with a native-$\chi^{(3)}$ resonance: when does cascading help?

The numerical estimate of the previous section – $|\chi^{(3)}_{\text{casc}}| \simeq 49\,|\chi^{(3)}_{\text{native}}|$ at $Q_2 = 4\times 10^3$ and $\Delta\omega = 2\gamma_2$ (Section 10) – begs the question: is the cascaded route actually preferable to engineering a single high-Q resonance that enhances the *native* bulk $\chi^{(3)}$? Both architectures use the same trick (a high-Q dielectric resonance at the fundamental) to build up intracavity intensity; they differ only in what mechanism inside the cavity then turns that intensity into a phase shift. The comparison is delicate because the two architectures have different bandwidth structures and different saturation channels, and the naive ratio $|\chi^{(3)}_{\text{casc}}|/|\chi^{(3)}_{\text{native}}|$ does not by itself settle the matter. We answer it here within the same TCMT framework.

Unified TCMT for both architectures

Consider a single FM cavity of frequency ω_1 and total decay γ_1, identical in both architectures. The TCMT equations (22)–(23) already include both candidate sources of $\chi^{(3)}_{\text{eff}}$ on the FM: the bulk-Kerr coefficient α_3 and the cascaded back-action via β and the SH mode a_2. The two architectures are limits of the same model:

- **Architecture A (native-$\chi^{(3)}$ resonance):** $\beta \to 0$ (no SH mode engineered, or $|\Delta\omega| \to \infty$). Only the FM resonance is used; the cubic on the FM is the bulk α_3.
- **Architecture B (cascaded $\chi^{(2)}$:$\chi^{(2)}$):** $\alpha_3 \to 0$ (work in a material with negligible native Kerr, or simply isolate the cascaded contribution from a full-Kerr model). All the action comes from β and the SH mode at $\omega_2 \approx 2\omega_1$.
- **Architecture A+B (combined):** both terms present; relevant for materials with appreciable $\chi^{(2)}$ *and* $\chi^{(3)}$.

Because the FM cavity is the same, the (linear-cavity, on-resonance) build-up factor $|a_1|^2/|s_+|^2 = 4\gamma_1^{\text{rad}}/\gamma_1^2$ (which equals $4/\gamma_1$ in the lossless / fully-radiative limit $\gamma_1^{\text{rad}} = \gamma_1$, and $2/\gamma_1$ at the conventional critical-coupling point $\gamma_1^{\text{rad}} = \gamma_1/2$ – the formulae below assume the former), and the interaction time $2/\gamma_1$ are identical. Any architectural advantage must therefore come from what multiplies these factors.

Per-photon nonlinear phase: the Q-scaling

At pump frequency $\omega_p = \omega_1$ and small signal (no saturation, no bistability) the steady-state intracavity intensity is $|a_1|^2 = (4/\gamma_1)\,|s_+|^2$, so the accumulated nonlinear phase $\phi_{\text{NL}} = \alpha_3^{\text{eff}}\,|a_1|^2 \cdot (2/\gamma_1)$ per input photon takes the simple form

$$\boxed{\frac{\phi_{\text{NL}}}{|s_+|^2} \;=\; \frac{8\,\alpha_3^{\text{eff}}}{\gamma_1^2}} \tag{42}$$

where α_3^{eff} is whichever Kerr coefficient drives the FM cavity. For the two architectures,

$$\text{Architecture A:} \qquad \alpha_3^{\text{eff,A}} = \alpha_3 \;\propto\; \chi^{(3)}_{\text{native}}, \tag{43}$$

$$\text{Architecture B:} \qquad \alpha_3^{\text{eff,B}} = -\frac{|\beta|^2}{\Delta\omega + i\gamma_2/2} \;\propto\; \frac{|\chi^{(2)}|^2\,\eta_{\text{geom}}}{n_2^2\,\Delta\omega} \;\propto\; Q_2. \tag{44}$$

The structural difference is condensed in one parameter: the *cascaded enhancement ratio*

$$\boxed{R \;\equiv\; \frac{|\alpha_3^{\text{eff,B}}|}{|\alpha_3^{\text{eff,A}}|} \;=\; \frac{|\beta|^2}{|\alpha_3|\,|\Delta\omega + i\gamma_2/2|} \;\overset{\Delta\omega \gg \gamma_2}{\simeq}\; \frac{|\chi^{(3)}_{\text{casc}}|}{|\chi^{(3)}_{\text{native}}|}\,.} \tag{45}$$

R is the proper figure of merit because it depends only on the material constants and the SH-resonance design (Q_2, $\Delta\omega$, the overlap η_{geom}), *not* on the FM cavity properties Q_1, γ_1^{rad} which both architectures share. Note that R as defined uses $|\alpha_3^{\text{eff,B}}|$ (the modulus, including $i\gamma_2/2$), while the asymptotic identification with $|\chi^{(3)}_{\text{casc}}|/|\chi^{(3)}_{\text{native}}|$ uses the real part of $\alpha_3^{\text{eff,B}}$ only; the two agree to within $\sim 3\%$ at $\Delta\omega = 2\gamma_2$ and to $\sim 11\%$ at $\Delta\omega = \gamma_2$.

Q-scaling. Combining (42) with (43)–(44):

$$\phi_{\mathrm{NL}}^{\mathrm{A}} \;\propto\; Q_1^2\,\chi^{(3)}_{\mathrm{native}}, \tag{46}$$

$$\phi_{\mathrm{NL}}^{\mathrm{B}} \;\propto\; Q_1^2\,Q_2\,\frac{|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}}{n_2^2}. \tag{47}$$

Both architectures benefit equally from Q_1 (the FM-cavity build-up scales as Q_1 and the interaction time scales as Q_1, giving the Q_1^2 factor common to both). The cascade adds an *extra* Q_2 factor: increasing the SH quality factor at fixed detuning ratio $\Delta\omega/\gamma_2$ pulls more action out of the same intracavity FM intensity.

The phase–bandwidth locus

Pushing either Q_1 or Q_2 to gain phase costs bandwidth. The useful operational bandwidth of each architecture is

$$\begin{aligned}&\text{Architecture A:} \quad \Delta f_{\mathrm{A}} = \gamma_1/(2\pi) = \omega_1/(2\pi Q_1),\\ &\text{Architecture B:} \quad \Delta f_{\mathrm{B}} = \min(\gamma_1,\gamma_2)/(2\pi).\end{aligned}$$

For architecture B, γ_2 enters because the cascaded coefficient itself has finite bandwidth (Section 7 sum rule); whichever of γ_1 or γ_2 is smaller bottlenecks the device.

Plotting $\phi_{\mathrm{NL}}/|s_+|^2$ versus Δf in log–log reveals three distinct families with three distinct slopes:

Family	Operating mode	Slope on log–log
A: native, sweep Q_1	$\phi \propto Q_1^2$, $\Delta f \propto 1/Q_1$	-2
$\mathrm{B_{II}}$: cascade, fix Q_1, sweep Q_2	$\phi \propto Q_2$, $\Delta f \propto 1/Q_2$	-1 (sum-rule)
$\mathrm{B_{I}}$: cascade, fix Q_2, sweep Q_1	$\phi \propto Q_1^2$, $\Delta f \propto 1/Q_1$	-2

The native locus A is a single line; the cascade is a two-parameter *region* in $(\phi, \Delta f)$ bounded by $\mathrm{B_I}$ and $\mathrm{B_{II}}$. Three qualitative features follow:

1. **Vertical offset.** At a given bandwidth, the cascade locus is shifted above the native locus by the structural ratio R of (45). $R \gg 1$ is the regime in which cascading wins outright; $R \lesssim 1$ is the regime in which native wins.

2. **Slopes differ.** A slope--1 line and a slope--2 line cross. For $R \gg 1$, the crossover happens at very narrow bandwidth – well past anything experimentally interesting, so cascade wins everywhere. For $R \sim 1$, the crossover sits in the experimentally relevant range and the architecture choice depends on the bandwidth target.

3. **Cascade beats native by an extra Q_1 if you can push Q_1.** Comparing slopes within the cascade region itself, family $\mathrm{B_I}$ (varying Q_1 at fixed Q_2) lies above $\mathrm{B_{II}}$ (varying Q_2 at fixed Q_1) in the high-Q_1 direction. To make this quantitative, define the phase-bandwidth product $\Pi \equiv \phi_{\mathrm{NL}} \cdot \Delta f$. At $Q_1 = Q_2 = Q$ (matched-Q corner, $\gamma_1 = \gamma_2$), using (46)–(47) and $\Delta f = \gamma_1/(2\pi) \propto 1/Q$:

$$\begin{aligned}\Pi^{\mathrm{A}}\big|_{Q_1=Q} \;&\propto\; \chi^{(3)}_{\mathrm{native}}\,Q^2 \cdot Q^{-1} \;=\; \chi^{(3)}_{\mathrm{native}}\,Q,\\ \Pi^{\mathrm{B}}\big|_{Q_1=Q_2=Q} \;&\propto\; \frac{|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}}{n_2^2}\,Q^2\,Q \cdot Q^{-1} \;=\; \frac{|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}}{n_2^2}\,Q^2.\end{aligned}$$

The cascade phase-bandwidth product at the matched-Q corner scales as $Q^2 \cdot R^{(0)}$, where $R^{(0)} \equiv |\chi^{(2)}|^2 \eta_{\text{geom}}/(n_2^2 \chi^{(3)}_{\text{native}})$ is the structural factor at unit Q_2. The native architecture scales only as Q. The cascade therefore beats native in phase-bandwidth by a factor of Q at the matched corner – one extra power of Q relative to the slope-only comparison.

The crossover bandwidth

Equating the native phase (46) to the cascade phase (47) at the bandwidth-matched point $\gamma_1 = \gamma_2$ (the SH bottleneck balances the FM bandwidth) and using the operating point $\Delta\omega = 2\gamma_2$ of Section 10 (so $|\chi^{(3)}_{\text{casc}}| \sim Q_2/(6\, n_2^2) \cdot |\chi^{(2)}|^2 \eta_{\text{geom}}$):

$$\chi^{(3)}_{\text{native}}\, Q_1^2 \;\sim\; \frac{|\chi^{(2)}|^2\, \eta_{\text{geom}}}{6\, n_2^2}\, Q_1^2\, Q_2 \;\Longrightarrow\; Q_2^{\text{crossover}} \;\sim\; \frac{6\, n_2^2\, \chi^{(3)}_{\text{native}}}{|\chi^{(2)}|^2\, \eta_{\text{geom}}}. \tag{48}$$

For $LiNbO_3$ at the values of Section 10, $Q_2^{\text{crossover}} \sim 80$: any SH cavity with Q_2 above this threshold tips the comparison in favour of the cascade. Since $Q_2 \sim 4000$ is routine in qBIC metasurfaces, $LiNbO_3$ *always* prefers cascade in this regime.

For AlGaAs near the band edge, native $\chi^{(3)}_{\text{native}} \sim 10^{-18}$ m^2/V^2 is enhanced by two-photon-resonance proximity while $|\chi^{(2)}| \sim 10^{-10}$ m/V is only modestly larger than $LiNbO_3$; (48) gives $Q_2^{\text{crossover}} \sim 2\times10^4$, which is above routinely achievable qBIC Q's, so native wins unless the SH cavity reaches that range.

For Si and Si_3N_4 the comparison is moot: $\chi^{(2)} \to 0$ by inversion symmetry, and cascading is simply not available. These materials have to rely on native-Kerr resonance (or move to periodically-poled cousin materials).

Material survey

Combining (45) with the explicit form (35) at the nominal design point ($Q_2 = 4\times10^3$, $\Delta\omega = 2\gamma_2$, $\eta_{\text{geom}} = 0.3$, n_2 from the material) gives the cascaded enhancement ratio for a few representative materials:

Material	$\lvert\chi^{(2)}\rvert$ [m/V]	$\lvert\chi^{(3)}_{\text{native}}\rvert$ [m^2/V^2]	n_2	R at Q_2=4×10^3	Q_2^{cross}
$LiNbO_3$ (telecom)	5×10^{-11}	2×10^{-21}	2.26	∼49	∼ 80
GaP (NIR)	1×10^{-10}	$\sim10^{-19}$	3.1	∼2	$\sim 2\times10^3$
AlGaAs (∼1550 nm)	1×10^{-10}	$\sim10^{-18}$	3.3	∼0.2	$\sim 2\times10^4$
AlN	1×10^{-11}	$\sim2\times10^{-21}$	2.1	∼2	$\sim 2\times10^3$
Si	$\equiv 0$	$\sim4\times10^{-18}$	3.5	— (no cascade)	
Si_3N_4	~ 0	$\sim2.5\times10^{-21}$	2.0	— (no cascade)	

Three regimes emerge. In materials where the native $\chi^{(3)}$ is small relative to $|\chi^{(2)}|^2/(n_2^2\omega_1)$ – chiefly $LiNbO_3$ – the cascade wins by well over an order of magnitude at routine Q_2 and increasing Q_2 buys further linear gain. In materials of intermediate $\chi^{(3)}_{\text{native}}$ such as GaP and AlN, the cascade is modestly ahead at $Q_2 \sim 4\times10^3$ ($R \sim 2$), and the margin grows linearly above the crossover (48). In materials where native $\chi^{(3)}$ is enhanced by electronic resonances near the band edge (AlGaAs), the comparison flips: native bulk-Kerr resonance is stronger unless Q_2 reaches $\sim 2\times10^4$, and the engineering simplicity of a single resonance is a free bonus. (GaAs, whose larger $|\chi^{(2)}| \simeq 350$ pm/V partly offsets its band-edge Kerr, sits near break-even, $R \simeq 1.4$ at $Q_2 = 4\times10^3$.) Si and Si_3N_4 have no cascade at all. (At the edge-of-adiabatic operating point $\Delta\omega \sim \gamma_2$ of Section 10, the R values above double; the qualitative picture is unchanged.)

Caveats specific to this comparison

1. *Cascade-specific loss.* The cascaded TPA-like coefficient $\gamma_3^{\text{eff}} = |\beta|^2\gamma_2/(2\Delta\omega^2)$ of Section 6(ii) has no analogue in low-loss native materials like $LiNbO_3$. At fixed phase shift, cascade always carries more loss than native, by a factor $\gamma_2/(2\Delta\omega)$ for the Kerr-to-loss ratio. This penalises cascade in all-optical-switching contrast and in low-noise applications.
2. *Lower saturation.* The cascade saturates at $|\beta||a_1| \sim |\Delta\omega|$, beyond which the trivial fixed point becomes parametrically unstable (Section 13, item 2). At fixed $\Delta\omega/\gamma_2$, the saturated FM amplitude $|a_1|_{\text{sat}} = |\Delta\omega|/|\beta| \propto \gamma_2 \propto 1/Q_2$ (because $|\Delta\omega| \propto \gamma_2 \propto 1/Q_2$ while $|\beta|$ is set by the modal overlap and is Q_2-independent), so the maximum useful stored energy $|a_1|^2_{\text{sat}} \propto 1/Q_2^2$. Native-$\chi^{(3)}$ saturation, by contrast, is set by thermal/damage limits typically orders of magnitude higher.
3. *Fabrication complexity.* Architecture A requires one high-Q resonance at ω_1; architecture B requires *two* high-Q resonances at ω_1 and $2\omega_1$ with simultaneously high spatial overlap. Disorder broadens γ_2 inhomogeneously and can wash out the cascade-enhancement long before it affects a single-resonance native design.
4. *Speed.* Native Kerr is essentially instantaneous ($\lesssim 10$ fs in dielectrics); cascade has bandwidth γ_2, i.e. ps-scale for routine Q_2. For ultrafast applications (sub-ps), only native-Kerr is fast enough; cascade is locked out by its own sum rule.

Summary of the comparison

Question	Answer	When it matters
At fixed BW, who wins?	Cascade by factor R; native if $R < 1$.	Sets the architecture decision for any narrowband application.
How does it scale with Q?	$\phi^{\text{A}} \propto Q_1^2$; $\phi^{\text{B}} \propto Q_1^2 Q_2$.	Cascade has a free extra Q-knob; useful in high-Q regime.
What's the price of more Q?	A: BW shrinks linearly; B: BW shrinks linearly in Q_1, also in Q_2.	Both pay the sum rule; cascade more so.
Saturation?	A: thermal/damage. B: $\lvert\beta a_1\rvert \sim \lvert\Delta\omega\rvert$, much lower.	Cascade is the limited one at high powers.
Ultrafast?	A: $\sim$ fs; B: $\sim 1/\gamma_2$ ps.	Use native for pulses $\lesssim 1$ ps.
Material choice?	A: any high-$\chi^{(3)}_{\text{native}}$ low-loss dielectric. B: needs $\chi^{(2)} \neq 0$.	$\chi^{(2)}$-active materials only for cascade.

The take-away: the cascade is the right choice for *narrowband, CW-driven nonlinear photonics in $\chi^{(2)}$-active materials* ($LiNbO_3$, GaP at telecom). For Si/Si_3N_4 photonics, ultrafast switching, high-power operation, or applications where two-photon loss is intolerable, a native-$\chi^{(3)}$ resonance is the right answer.

12 Application landscape: Kerr-driven observables and devices

The previous section reduced the architecture question to a single structural ratio $R \equiv |\chi^{(3)}_{\text{casc}}|/|\chi^{(3)}_{\text{native}}|$. But R is only directly observable through a specific class of measurements: those that depend on the FM-cavity Kerr coefficient $\alpha_3^{\text{eff}} = \alpha_3 + \alpha_3^{\text{casc}}$. Different Kerr-driven observables scale differently with α_3^{eff} – linearly, inversely, or quadratically – and the cascade therefore appears in

each at a different "leverage". In addition, several capabilities exist that have *no* native-Kerr analogue at all, and these are where the cascaded route is uniquely useful rather than merely quantitatively better.

We organise the discussion in three tiers. Tier 1 collects observables that admit a clean architecture-vs-architecture comparison of the form (A+B)/(A)= $f(R)$. Tier 2 collects capabilities specific to the cascade with no native counterpart. Tier 3 lists applications that lie outside the single-cavity TCMT framework and require extensions we do not develop here.

12.1 Tier 1: observables with quantitative R-scaling

1. Self-phase modulation (SPM). Already established in Section 11. On-resonance pump at $\omega_p = \omega_1$ in the fully-radiative limit $\gamma_1^{(\mathrm{rad})} = \gamma_1$ gives a nonlinear phase $\phi_{\mathrm{NL}} = 8\alpha_3^{\mathrm{eff}}|s_+|^2/\gamma_1^2$ per input photon (42) (at the conventional critical-coupling point $\gamma_1^{(\mathrm{rad})} = \gamma_1/2$, $|a_1|^2 = 2|s_+|^2/\gamma_1$ and $\phi_{\mathrm{NL}} = 4\alpha_3^{\mathrm{eff}}|s_+|^2/\gamma_1^2$; we quote the lossless value to match (42)). Since ϕ_{NL} is linear in $\alpha_3^{\mathrm{eff}} = \alpha_3 + \alpha_3^{\mathrm{casc}}$, and using the cascaded ratio (45), the architecture comparison gives

$$\boxed{\frac{\phi_{\mathrm{NL}}^{\mathrm{A+B}}}{\phi_{\mathrm{NL}}^{\mathrm{A}}} = \frac{\alpha_3 + \alpha_3^{\mathrm{casc}}}{\alpha_3} = 1 + R\,.} \tag{49}$$

Linear in R; the foundational observable. Read off as intensity-dependent transmission lineshape distortion or via heterodyne phase measurement at the FM port.

2. Cross-phase modulation (XPM). A separate weak signal beam, co-resonant with the FM cavity but distinguishable by polarisation or by frequency offset $|\Omega| < \gamma_1$, experiences a refractive-index shift driven by the pump intensity $|a_1^{\mathrm{pump}}|^2$. In the cascade picture, the pump generates an SH field that mediates DFG mixing onto the signal; in the native picture, the bulk $\chi^{(3)}$ provides the mixing directly. Both channels carry the same Boyd degeneracy factor of 6 (the index-permutation factor $2(2-\delta_{ij})$ for distinct fields), so the relative architecture ratio inherits the SPM result:

$$\boxed{\frac{\alpha_3^{\mathrm{XPM,\,A+B}}}{\alpha_3^{\mathrm{XPM,\,A}}} = 1 + R\,, \qquad \alpha_3^{\mathrm{XPM}} = 2\,\alpha_3^{\mathrm{SPM}}\,.} \tag{50}$$

The factor 2 between XPM and SPM holds independently in both architectures. Experimentally: pump-probe transmission, with the probe phase shift versus pump intensity read out via interferometry or heterodyne.

Important restriction. XPM in the strict sense requires a second beam (distinguishable from the pump). A single beam at ω_p with a weak modulation at $\omega_p \pm \Omega$ instead drives *four-wave mixing* (item 4 below), not XPM, because the cascade generates the conjugate sideband through DFG rather than a phase shift. The XPM coefficient as defined here applies to a genuinely separate signal channel.

Tensor-channel caveat. The $1+R$ ratio holds in the *scalar* TCMT picture used here. In a real metasurface, the cascade XPM goes through pump→SH→DFG-with-signal, which involves $\chi^{(2)}_{ijk}$ contracted with mixed polarisation indices ($e{\cdot}e{\to}$SH, then SH$\cdot o{\to}$signal for a $3m$ crystal with e-polarised pump and o-polarised signal, for example). These tensor contractions can differ from the $\chi^{(2)}_{eee}$ that mediates SHG of the pump alone – in some polarisation channels they vanish by selection rule, in others they are comparable. The native-XPM channel by contrast uses $\chi^{(3)}_{eooe}$ or similar, which is governed by the host $\chi^{(3)}$ tensor. The $1+R$ ratio holds for tensor channels in which both cascade and native pieces are non-vanishing and geometrically similar; in mixed-channel experiments the relative factor may differ. A practical experiment should verify the relevant tensor contraction for the intended polarisation geometry.

3. Bistability threshold and all-optical switching energy. The steady-state amplitude equation at pump detuning $\delta_p \equiv \omega_p - \omega_1$, obtained by setting $\dot{a}_1 = 0$ in (23) (with a_2 adiabatically eliminated and the cascaded TPA neglected for the cusp analysis), is

$$\left[\tfrac{\gamma_1}{2} - i\big(\delta_p + \alpha_3^{\text{eff}}|a_1|^2\big)\right] a_1 \;=\; \sqrt{\gamma_1^{\text{rad}}}\, s_+. \tag{51}$$

Taking moduli squared and writing $y \equiv \alpha_3^{\text{eff}}|a_1|^2/(\gamma_1/2)$, $x \equiv \delta_p/(\gamma_1/2)$, $u \equiv |\alpha_3^{\text{eff}}|\,|s_+|^2\,\gamma_1^{\text{rad}}/(\gamma_1/2)^3$ (a dimensionless input intensity), (51) becomes the standard Kerr-bistability cubic $y[1 + (x - y)^2] = u$. This cubic has three real roots $y_- < y_{\text{mid}} < y_+$ – the bistable regime – iff the discriminant condition is satisfied, which is well known [21, 5] to require

$$\text{sign}(\delta_p) = -\text{sign}(\alpha_3^{\text{eff}}), \qquad \frac{|\delta_p|}{\gamma_1/2} > \sqrt{3} \iff |\delta_p| > \frac{\sqrt{3}}{2}\,\gamma_1. \tag{52}$$

The sign condition is automatic: a positive (self-focusing) $\alpha_3^{\text{eff}} > 0$ pulls the cavity resonance down with growing intensity, so a red-detuned pump ($\delta_p < 0$) gets pulled into resonance and bistability ensues; for a self-defocusing $\alpha_3^{\text{eff}} < 0$ (e.g. a cascade with $\Delta\omega > 0$), the same analysis with the opposite sign of δ_p applies. The $|\delta_p| > \sqrt{3}\gamma_1/2$ inequality is the cusp condition.

The snap-up power follows from the saddle-node bifurcation of the cubic. At the cusp $|\delta_p| = \sqrt{3}\gamma_1/2$, the critical intracavity intensity is $|a_1|_{\text{c}}^2 = |\delta_p|/|\alpha_3^{\text{eff}}| = \sqrt{3}\gamma_1/(2|\alpha_3^{\text{eff}}|)$, and the corresponding input power is

$$|s_+|^2_{\text{bist}} \;\propto\; \frac{\gamma_1^2\,|\delta_p|}{|\alpha_3^{\text{eff}}|\,\gamma_1^{\text{rad}}} \;\propto\; \frac{\gamma_1^2}{|\alpha_3^{\text{eff}}|} \quad \text{(at fixed } \delta_p/\gamma_1\text{)}. \tag{53}$$

Inversely proportional to the Kerr coefficient, so the architecture ratio is inverted relative to the phase shift:

$$\boxed{\frac{|s_+|^{2,\,\text{A+B}}_{\text{bist}}}{|s_+|^{2,\,\text{A}}_{\text{bist}}} = \frac{1}{1+R}\,.} \tag{54}$$

For $LiNbO_3$ ($R \sim 49$ at the nominal $\Delta\omega = 2\gamma_2$, Section 10), the cascade lowers the bistability threshold by $\sim 50\times$ (and by $\sim 100\times$ at the edge-of-adiabatic $\Delta\omega \sim \gamma_2$). Re-framed as an all-optical switching energy: biasing the system on the lower branch and applying a control pulse to push past the saddle-node, the minimum required control energy obeys

$$\boxed{\frac{E^{\text{A+B}}_{\text{switch}}}{E^{\text{A}}_{\text{switch}}} = \frac{1}{1+R}\,.} \tag{55}$$

The switching speed is set by the FM ringdown $1/\gamma_1$ (ps for $Q_1 = 10^3$); the readout is a direct power measurement on a photodiode. **This is the canonical photonic-computing application.**

4. Four-wave-mixing (FWM) wavelength conversion. Driving with a strong pump at $\omega_p \approx \omega_1$ and a weak signal at $\omega_s = \omega_p - \Omega$ generates an idler at $\omega_i = 2\omega_p - \omega_s = \omega_p + \Omega$. In the cascade mechanism: the pump generates SH at $2\omega_p \approx \omega_2$, which then DFG-mixes with the signal to produce the idler at the FM cavity.

Derivation of the conversion efficiency. Write the FM amplitude as $a_1 = a_p + a_s\,e^{-i\Omega t} + a_i\,e^{+i\Omega t}$ in the frame rotating at ω_p, with $|a_s|, |a_i| \ll |a_p|$. Substituting into the effective FM equation $\dot{a}_1 = (-\gamma_1/2)\,a_1 + i\,\alpha_3^{\text{eff}}|a_1|^2 a_1 + \sqrt{\gamma_1^{\text{rad}}}s_+$ and collecting the $e^{+i\Omega t}$ Fourier component generates an idler source

$$\mathcal{S}_i \;=\; i\,\alpha_3^{\text{eff}}\,a_p^2\,a_s^*,$$

which is the standard FWM source proportional to pump$^2\times$ signal-conjugate. Expanding $|a_1|^2 a_1$ to first order in the small fields and collecting the $e^{+i\Omega t}$ Fourier component gives two pieces: $a_p^2\, a_s^*$ (the FWM source above) and a $2|a_p|^2 a_i$ piece that renormalises the idler detuning – a pump-induced XPM self-shift that adds a real part $\alpha_3^{\rm eff}\, 2|a_p|^2$ to the idler resonance condition. In the small-signal estimate below we drop this self-shift (it is $O(\alpha_3^{\rm eff}|a_p|^2/\gamma_1)$ smaller than the propagator denominator $\gamma_1/2$ once the pump is below the bistability threshold of item 3). The idler steady-state response is then

$$a_i \;=\; \frac{i\,\alpha_3^{\rm eff}\, a_p^2\, a_s^*}{\gamma_1/2 - i\Omega} \;\overset{|\Omega|<\gamma_1/2}{\simeq}\; \frac{i\,\alpha_3^{\rm eff}\, a_p^2\, a_s^*}{\gamma_1/2},$$

and the idler output via the input–output relation $s_-^{(i)} = \sqrt{\gamma_1^{\rm rad}}\, a_i$ gives an idler power $|s_-^{(i)}|^2 = \gamma_1^{\rm rad}|a_i|^2$ while the signal input generates an intracavity signal $|a_s| = 2\sqrt{\gamma_1^{\rm rad}}|s_+^{(s)}|/\gamma_1$. Combining,

$$\eta_{\rm FWM} \;\equiv\; \frac{|s_-^{(i)}|^2}{|s_+^{(s)}|^2} \;\propto\; \frac{|\alpha_3^{\rm eff}|^2\,|a_p|^4}{(\gamma_1/2)^2}\,, \tag{56}$$

quadratic in the effective Kerr coefficient. The architecture ratio is therefore

$$\boxed{\frac{\eta_{\rm FWM}^{\rm A+B}}{\eta_{\rm FWM}^{\rm A}} = (1+R)^2\,.} \tag{57}$$

For $LiNbO_3$ ($R\sim 49$ at the nominal $\Delta\omega = 2\gamma_2$), this is a factor of $\sim 2.5\times 10^3$ in conversion efficiency at the conservative operating point, and approaches $\sim 10^4$ if one operates at $\Delta\omega\sim\gamma_2$. Either way, FWM is the most strongly differentiating single-observable test of the cascade mechanism.

Bandwidth constraint. All three frequencies must lie within $\min(\gamma_1,\gamma_2)$. For $Q_1 \ll Q_2$ the FM linewidth bounds Ω via the $1/(\gamma_1/2 - i\Omega)$ factor in a_i; for $Q_1 \gtrsim Q_2$ the cascade bandwidth γ_2 takes over via the AC response (37) of $\alpha_3^{\rm eff}(\Omega)$. Cascade-FWM is therefore restricted to a frequency window $\Omega \lesssim \min(\gamma_1,\gamma_2)/(2\pi) \sim$ GHz at telecom Q's.

12.2 Tier 2: cascade-only capabilities

These applications exploit aspects of the cascaded $\chi_{\rm eff}^{(3)}$ that have no analogue in a native-$\chi^{(3)}$ resonance: the $\Delta\omega$-dependence of the coefficient itself, and the imaginary part $\gamma_3^{\rm eff} = |\beta|^2(\gamma_2/2)/[\Delta\omega_{\rm SH}^2 + (\gamma_2/2)^2]$ which produces an intensity-dependent loss whose magnitude is *also* $\Delta\omega$-tunable. Architecture A simply does not provide either knob.

5. Sign-tunable saturable absorber. Throughout this item we write $\Delta\omega_{\rm SH} \equiv \Delta\omega = 2\omega_1 - \omega_2$ for the SH detuning, to distinguish it visually from the FM pump detuning δ_p that appears in items 3 and 4 above; numerically they are the same quantity used since §5.

Operating near the SH resonance ($|\Delta\omega_{\rm SH}| \lesssim \gamma_2$), take the imaginary part of (20). Writing $1/(\Delta\omega + i\gamma_2/2) = (\Delta\omega - i\gamma_2/2)/[\Delta\omega^2 + (\gamma_2/2)^2]$, the imaginary part of $\chi_{\rm casc}^{(3)}$ is

$$\mathrm{Im}\,\chi_{\rm casc}^{(3)}(\Delta\omega) \;=\; \frac{2\,\varepsilon_0\,\omega_1}{3\,N_2}\cdot\frac{|M|^2\,(\gamma_2/2)}{[\Delta\omega^2 + (\gamma_2/2)^2]\int|\mathbf{f}_1|^4\,d^3r}. \tag{58}$$

The corresponding TCMT loss coefficient (obtainable directly by keeping the imaginary part of the adiabatic-elimination step in Section 6 without expanding for small $\gamma_2/\Delta\omega$) is

$$\gamma_3^{\rm eff}(\Delta\omega_{\rm SH}) \;=\; \frac{|\beta|^2\,(\gamma_2/2)}{\Delta\omega_{\rm SH}^2 + (\gamma_2/2)^2}, \tag{59}$$

which is precisely $\gamma_3^{\rm eff} = |\beta|^2\gamma_2/(2\Delta\omega^2)$ of Section 6(ii) in the adiabatic limit $|\Delta\omega| \gg \gamma_2$, and rises to a Lorentzian peak at $\Delta\omega_{\rm SH} = 0$:

$$\gamma_3^{\rm eff}(\Delta\omega_{\rm SH}=0) \;=\; \frac{|\beta|^2}{\gamma_2/2} \;=\; \frac{2\,Q_2}{\omega_2}\,|\beta|^2 \qquad \text{(Lorentzian, FWHM} = \gamma_2\text{)}. \tag{60}$$

In the language of an effective material parameter, using (58) and the same $1/n_2^2$ analysis as in Section 6 (replacing $\Delta\omega$ by $\gamma_2/2$ in the denominator of the explicit form (35)),

$$\mathrm{Im}\,\chi^{(3)}_{\rm casc}\big|_{\Delta\omega_{\rm SH}=0} \;\sim\; \frac{4\,\omega_1}{3\,n_2^2\,\gamma_2}\,|\chi^{(2)}|^2\,\eta_{\rm geom} \;=\; \frac{2\,Q_2}{3\,n_2^2}\,|\chi^{(2)}|^2\,\eta_{\rm geom}, \tag{61}$$

where we used $\omega_1/\gamma_2 = Q_2/2$. Compared with the real part at $\Delta\omega = \gamma_2/2$ (where $|\mathrm{Re}\,\chi^{(3)}_{\rm casc}| = Q_2/(3n_2^2)\,|\chi^{(2)}|^2\eta_{\rm geom}$ from (35)), the peak imaginary part is a factor of 2 larger – the same Lorentzian peak-to-shoulder ratio that maps the dispersive real part to the absorptive imaginary part. This is a *saturable absorption* channel (intensity-dependent loss) that, unlike intrinsic two-photon absorption in semiconductors, is *detunable* away from resonance: shifting ω_2 by $\gtrsim$ a few γ_2 (electro-optically, thermo-optically, or by strain) suppresses the loss by $1/\Delta\omega_{\rm SH}^2$ from (59).

Comparison with SESAM (semiconductor saturable absorber mirror, the workhorse of pulsed-laser mode locking [22]): SESAMs have a fixed, material-determined recovery time set by excited-state population dynamics; cascaded SA has recovery time $\sim 1/\gamma_2$ (ps for $Q_2 = 4{\times}10^3$) and is all-optical, with no excited-state physics or thermal lifetime drift. The trade-off is that cascaded SA requires a $\chi^{(2)}$-active substrate and double-resonance fabrication.

6. All-optical amplitude modulator via $\Delta\omega$ tuning. Modulating $\Delta\omega(t) = \Delta\omega_0 + \delta\Omega\cos(\Omega_{\rm mod}t)$ via electro-optic, thermo-optic, or piezo-electric actuation of the SH cavity periodically modulates $\alpha_3^{\rm eff}$, and therefore the FM transmission $|s_-/s_+|^2$ at fixed pump. For small modulation $\delta\Omega \ll |\Delta\omega_0|$ and $|\Delta\omega_0| \gg \gamma_2/2$, the adiabatic-limit derivative

$$\left.\frac{\partial\alpha_3^{\rm casc}}{\partial\Delta\omega}\right|_{\Delta\omega_0} \;\simeq\; \frac{|\beta|^2}{\Delta\omega_0^2} \tag{62}$$

gives, in the DC limit ($\Omega_{\rm mod} \to 0$), a fractional modulation $\delta\alpha_3^{\rm casc}/\alpha_3^{\rm casc} = \delta\Omega/\Delta\omega_0$: a fractional SH-cavity detuning produces an equal fractional change in $\alpha_3^{\rm casc}$.

Transfer function: low-pass plus AC band-pass. For finite $\Omega_{\rm mod}$ the cascaded coefficient cannot track the SH-cavity detuning instantaneously. The correct response must be obtained from the *time-dependent* SH equation with modulated detuning – linearising $\Delta\omega(t) = \Delta\omega_0 + \delta\Omega\cos(\Omega_{\rm mod}t)$ in $\delta\Omega$ and solving the SH equation in steady state gives, for the *positive-frequency sideband* of δa_2,

$$A_+(\Omega_{\rm mod}) \;=\; \frac{(\delta\Omega/2)\,a_{2,0}}{\gamma_2/2 + i(\Omega_{\rm mod} - \Delta\omega_0)}, \qquad a_{2,0} = \frac{-\beta\,a_1^2}{\Delta\omega_0 + i\gamma_2/2}, \tag{63}$$

which is resonant at $\Omega_{\rm mod} = \Delta\omega_0$ (the sideband of $\Delta\omega(t)$ matches the SH-cavity resonance in the rotating frame). The fractional a_2 modulation, which carries directly into the cascade contribution to $\alpha_3^{\rm eff}$, is

$$\left|\frac{\delta a_2}{a_{2,0}}\right|_{\Omega_{\rm mod}} \;=\; \frac{\delta\Omega/2}{\sqrt{(\gamma_2/2)^2 + (\Omega_{\rm mod} - \Delta\omega_0)^2}}. \tag{64}$$

Its values at the two limits are

$$\left|\frac{\delta a_2}{a_{2,0}}\right|_{\text{peak at}\;\Omega = \Delta\omega_0} \;=\; \frac{\delta\Omega}{\gamma_2}, \qquad \left|\frac{\delta a_2}{a_{2,0}}\right|_{\rm DC} \;=\; \frac{\delta\Omega}{\sqrt{\Delta\omega_0^2 + (\gamma_2/2)^2}}, \tag{65}$$

so the AC-peak enhancement over DC is

$$\boxed{\frac{|\delta\alpha_3^{\rm eff}|_{\text{peak at }\Omega_{\rm mod}=\Delta\omega_0}}{|\delta\alpha_3^{\rm eff}|_{\rm DC}} = \frac{\sqrt{\Delta\omega_0^2+(\gamma_2/2)^2}}{\gamma_2} \xrightarrow[\Delta\omega_0\gg\gamma_2]{} \frac{|\Delta\omega_0|}{\gamma_2}.} \tag{66}$$

At the conservative operating point $\Delta\omega_0 = 2\gamma_2$ the AC peak is $\sqrt{4.25} \approx 2.06\times$ the DC modulation depth – a modest but useful boost. This number agrees with the TCMT measurement ($1.92\times$ in the AM-modulator figure of the simulation report). Note that this is the boost on the cascade-side $\alpha_3^{\rm eff}$ modulation amplitude; the FM-port transmission modulation also picks up a factor $1/|\gamma_1/2 + i\Omega_{\rm mod}|$ from the FM-cavity filtering, which is close to unity at $\Omega_{\rm mod} = 2\gamma_2$ when $\gamma_1 \gg 2\gamma_2$ (the worked-example regime).

The transfer function from $\delta\Omega(t)$ to the FM transmission modulation therefore has two features:

- a low-pass response with roll-off at $\Omega_{\rm mod} \sim \gamma_2/2$ (single-pole, set by the cascade bandwidth);
- a Lorentzian band-pass peak at $\Omega_{\rm mod} = \Delta\omega_0$ of FWHM γ_2 and peak-to-DC ratio $4\Delta\omega_0^2/\gamma_2^2$.

At telecom $Q_2 = 4\times10^3$ this gives a DC bandwidth $\lesssim \gamma_2/(2\pi) \simeq 97$ GHz and an AC operating point at $\Omega_{\rm mod}/(2\pi) = |\Delta\omega_0|/(2\pi) \simeq 200$ GHz when $\Delta\omega_0 = 2\gamma_2$.

This is a genuinely novel capability: native-Kerr cavities have no mechanism for converting a detuning modulation into an amplitude modulation, since the bulk $\chi^{(3)}$ is not detuning-dependent.

7. Passive optical limiter via the parametric ceiling. The cascade saturation $|\beta||a_1| \to |\Delta\omega|$ (Section 13, item 2) is more than a small-signal-breakdown caveat: it is a *hard, passive intensity clamp* at intracavity energy $|a_1|^2_{\rm cap} \simeq (\Delta\omega/\beta)^2$. Above this threshold the trivial fixed point destabilises and the FM clamps at the cap regardless of further pump increase, with the excess input dissipated through the cascaded TPA channel. In an integrated metasurface this is an all-optical *power limiter* – the same function that absorption-based limiters serve in laser safety glasses, but realised dispersively with no thermal load on the absorbing layer (loss is in the SH cavity, which can be heat-sunk separately). The ceiling is tunable by adjusting $\Delta\omega$: at the design point $\Delta\omega = 2\gamma_2$ the clamp is set; sweeping $\Delta\omega$ tunes the ceiling quadratically. Native-$\chi^{(3)}$ resonance has no such ceiling – intensity scales with input up to material damage. See §6–§13 for the underlying mechanism.

8. $\Delta\omega$-readout transducer for sensing. Inverting the previous: any process that shifts ω_2 (temperature, strain, applied electric field via Pockels effect on the SH cavity, adsorbed molecules, etc.) shifts $\Delta\omega$, which shifts $\alpha_3^{\rm eff}$, which shifts the SPM phase $\phi_{\rm NL}$. The cascade thus provides a built-in transduction from arbitrary ω_2-perturbations to a Kerr observable.

The sensitivity follows from the chain rule $d\phi_{\rm NL}/d\Delta\omega = (8|s_+|^2/\gamma_1^2)\cdot(d\alpha_3^{\rm casc}/d\Delta\omega)$ (using $\phi_{\rm NL} \propto \alpha_3^{\rm eff}$ from (42)). Differentiating the Lorentzian $\alpha_3^{\rm casc}(\Delta\omega) = -|\beta|^2\Delta\omega/[\Delta\omega^2 + (\gamma_2/2)^2]$ of Section 6 explicitly (with $b \equiv \gamma_2/2$):

$$\frac{d\alpha_3^{\rm casc}}{d\Delta\omega} = -|\beta|^2\,\frac{b^2-\Delta\omega^2}{[\Delta\omega^2+b^2]^2}\,. \tag{67}$$

Solving $d|d\alpha/d\Delta\omega|/d\Delta\omega = 0$ gives *two* extrema: a global maximum at $\Delta\omega = 0$ and a smaller local maximum at $|\Delta\omega| = b\sqrt{3}$ (factor of 8 below the global one). The peak slope is therefore

$$\boxed{\left|\frac{d\alpha_3^{\rm casc}}{d\Delta\omega}\right|_{\Delta\omega=0} = \frac{|\beta|^2}{b^2} = \frac{4\,|\beta|^2}{\gamma_2^2} \propto Q_2^2\,,} \tag{68}$$

giving

$$\left|\frac{d\phi_{\mathrm{NL}}}{d\Delta\omega}\right|_{\max} = \frac{32\,|s_+|^2\,|\beta|^2}{\gamma_1^2\,\gamma_2^2} \propto Q_1^2\,Q_2^2\,. \tag{69}$$

The sensor sensitivity scales as Q_2^2, not Q_2 – quadratically more sensitive in the SH cavity factor than the SPM phase shift itself, and gated by the FM cavity Q_1^2 in the usual way. The extra power of Q_2 comes from the Lorentzian slope: a coefficient of amplitude $\sim |\beta|^2/\gamma_2$ varies over a width $\sim \gamma_2$, giving a derivative $\sim |\beta|^2/\gamma_2^2$.

Operating-point caveat. $\Delta\omega = 0$ is also exactly where the cascaded two-photon-like loss $\gamma_3^{\mathrm{eff}} = |\beta|^2 b/[\Delta\omega^2 + b^2]$ peaks (eq. (59)) – the sensor's most sensitive point is its lossiest point. Useful figures of merit therefore depend on the weighting: the ratio (slope)$/\gamma_3^{\mathrm{eff}}$ is monotonic in $|\Delta\omega|$ within each Lorentzian branch and asymptotes to $2/\gamma_2$ at both $\Delta\omega \to 0$ and $|\Delta\omega| \to \infty$; it has a minimum (zero) at $|\Delta\omega| = b$ where the slope itself vanishes. In practice the sensor should be biased a few γ_2 off-resonance – enough to suppress TPA by $\sim (\gamma_2/\Delta\omega)^2$ while accepting a slope smaller than (68) by a similar factor. The fundamental Q_2^2 scaling survives this trade-off; the prefactor is reduced by an $O(1)$ amount depending on the chosen $\Delta\omega/\gamma_2$.

Shot-noise-limited figure of merit. The Q_2^2 scaling applies to the raw slope $d\alpha_3^{\mathrm{casc}}/d\Delta\omega$. For a shot-noise-limited readout the relevant figure of merit is slope$/\sqrt{\gamma_3^{\mathrm{eff}}}$ (the loss sets the noise floor through the detected-photon shot noise), which scales as $Q_2/\sqrt{Q_2} = Q_2^{3/2}$. Either scaling is faster than the linear Q_2 enhancement of the cascaded Kerr coefficient itself. Pushing Q_2 continues to pay off, but the practical FoM exponent depends on which noise channel dominates.

For a representative SH-cavity transducer in $LiNbO_3$ ($dn_e/dT \simeq 4\times10^{-5}$/K [23] gives a bare $d\omega_2/dT \simeq -\omega_2\,(dn/dT)/n \sim 2\pi \times 7$ GHz/K; the meta-atom-mode coupling reduces this by the mode-volume overlap with the $LiNbO_3$, typically a factor $\sim$ 3–5, giving an effective $\sim 2\pi \times$ 1–2 GHz/K), with $\gamma_2/2\pi \sim 100$ GHz at $Q_2 = 4\times10^3$, even after the TPA-penalised bias this resolves mK-scale temperature shifts at modest pump power and standard photodetector noise. Applications include refractometric biosensing, strain mapping, integrated thermometry, and (with appropriate cavity design) electric-field sensing via the Pockels effect.

This may be the most useful application of the cascade framework, because it converts a familiar Kerr-readout chain into a sensor whose figure of merit scales with Q_2^2 rather than the single-power-of-Q ceiling familiar from linear-cavity sensors.

12.3 Tier 3: applications outside the present scope

The following applications use the cascade but require theoretical extensions we do not develop:

- *Pulse compression and GVD compensation.* The sign-tunable cascaded Kerr can compensate either sign of group-velocity dispersion in subsequent free-space or fibre propagation, leading to soliton-like compression or anti-soliton chirp. This is a *propagation-level* effect; the metasurface provides the local nonlinearity, but characterisation requires convolution with downstream linear optics. See [9, 4].
- *Optical memory / flip-flop.* Same physics as all-optical switching (item 3 above); the "memory time" is the upper-branch dwell time, limited by thermo-optic drift, fabrication detuning, and slow background fluctuations rather than by anything in the TCMT model. Not a single-cavity TCMT effect to formalise.
- *Logic gates.* Two co-resonant control inputs driving one bistable element implement AND/OR by threshold combinations. This is a multi-port extension of bistability and requires the multi-port input-output theory of [7] rather than the single-port TCMT used here.

- *Degenerate optical parametric oscillation (OPO) and squeezed-light generation.* Above the parametric threshold the cascade system *is* an OPO, with above-threshold conversion efficiency and below-threshold squeezing of the appropriate quadrature. But OPO operation requires substantial $\gamma_2^{(\mathrm{rad})}$ at the SH port (to extract the parametric output), which contradicts the cascade design choice of suppressing SH out-coupling. These are therefore *different devices* with conflicting design constraints rather than alternative uses of the same metasurface. Squeezed light additionally requires quantum input-output theory beyond the classical TCMT framework adopted in this note.
- *Kerr frequency combs.* Single-mode cascaded $\chi^{(3)}$ can in principle support a Lugiato–Lefever-type comb, but this requires a multi-mode generalisation of the TCMT (one resonance per comb tooth) together with dispersion engineering of the FM family. A substantial separate project.
- *Phase conjugation.* The DFG step in the cascade is by construction phase-conjugate to the FM input. This is implicit in the dynamics of a_2 and the back-action onto a_1; making it explicit requires tracking the phase of the SH amplitude through the cascade and is best handled in a dedicated treatment.

12.4 Summary of observables

Observable	(A+B)/(A) ratio	Comment
SPM phase shift	$1+R$	Foundational
XPM phase shift	$1+R$	Same R; requires separate signal
Bistability threshold $\lvert s_+\rvert^2$	$1/(1+R)$	Inverted scaling
All-optical switching energy	$1/(1+R)$	Photonic-computing application
FWM conversion efficiency	$(1+R)^2$	Most differentiating
Saturable absorber rate	n/a	Cascade-only; $\Delta\omega$-tunable
AM modulator response	n/a	Cascade-only; $\Delta\omega(t)$-driven
Optical-limiter clamp $\lvert a_1\rvert^2_{\mathrm{cap}}$	$\sim(\Delta\omega/\beta)^2$	Cascade-only; passive intensity ceiling
Sensor sensitivity	$\propto Q_2^2$	Cascade-only; $\Delta\omega$-readout, biased a few γ_2 off-resonance

Tier 1 establishes the cascade as a quantitative replacement for bulk-Kerr-driven nonlinear optics, with leverage factors of $1+R$ to $(1+R)^2$ depending on observable. Tier 2 establishes the cascade as a source of *new* capabilities – saturable absorption, amplitude modulation, intensity limiting, and sensing – that the native-$\chi^{(3)}$ architecture cannot provide at any Q_1. Together the two tiers cover most of the experimental opportunities a doubly-resonant cascaded $\chi^{(3)}$ metasurface opens up.

13 Validity, caveats, and what to verify

1. **Adiabatic-elimination breakdown.** The naive form (33) ($\chi^{(3)}_{\mathrm{casc}} \propto 1/\Delta\omega$) diverges as $\Delta\omega \to 0$; the full Lorentzian $-|\beta|^2/(\Delta\omega + i\gamma_2/2)$ retained in (20) does not. The cascaded-Kerr description (*adiabatic* elimination of a_2) is strictly valid for $|\Delta\omega| \gtrsim 2\gamma_2$; we quote results down to $|\Delta\omega| \sim \gamma_2$ as an extrapolation that doubles the enhancement (real-part Lorentzian peaks at $|\Delta\omega| = \gamma_2/2$, so R is bounded above by twice its $\Delta\omega = 2\gamma_2$ value).

At $|\Delta\omega| = \gamma_2$ the loss-to-Kerr ratio $\gamma_3^{\text{eff}}/|\alpha_3^{\text{eff,casc}}| = \gamma_2/(2|\Delta\omega|)$ reaches 1/2 and the slow-FM-modulation ($|\dot{a}_1| \ll |\Delta\omega|\, a_1$) requirement also tightens; below $|\Delta\omega| \sim \gamma_2$ one must keep a_2 as a dynamical variable. On resonance ($\Delta\omega = 0$) the steady-state SH amplitude is bounded by $|a_2| = |\beta||a_1|^2/(\gamma_2/2)$.

2. **Pump depletion / saturation.** The derivation is small-signal: it assumes $|a_2| \ll |a_1|$, equivalently $|\beta||a_1|^2/|\Delta\omega| \ll |a_1|$, i.e. $|\beta||a_1| \ll |\Delta\omega|$. Above this threshold the SH is no longer slaved to the FM and the response saturates; in fact, exactly at the saturation point one finds an analogue of the parametric oscillation threshold. The maximum useful FM amplitude is therefore $|a_1|_{\text{sat}} = |\Delta\omega|/|\beta|$, which scales as $1/Q_2$ at fixed $\Delta\omega/\gamma_2$ (because $|\Delta\omega| \propto \gamma_2 \propto 1/Q_2$ while $|\beta|$ is set by the modal overlap, independent of Q_2); equivalently, the maximum stored energy $|a_1|^2_{\text{sat}} \propto 1/Q_2^2$. High-$Q_2$ devices saturate at lower powers.

3. **Two-photon-like loss.** The imaginary part of α_3^{eff} – the cascaded two-photon-absorption analogue – is unavoidable and bounded below by the same overlap. It does *not* dump energy outside the system in the limit $\gamma_2^{\text{rad}} \to 0$, but rather absorbs into the SH mode and from there into the material. This is identical to two-photon absorption from an experimental standpoint and limits the Kerr-driven phase shift before bleaching.

4. **Thermo-optic and free-carrier nonlinearities** in dielectric metasurfaces are notoriously easy to confuse with "Kerr". At the field intensities required to see a measurable cascaded $\chi^{(3)}$ phase shift (several MW/cm^2 in the resonator), thermal effects in absorbing materials (Si, GaAs near the band edge) often dominate. $LiNbO_3$ is comparatively benign in this respect.

5. **Tensor-overlap evaluation.** The dimensionless overlap η_{geom} is assumption-driven in this note. It must be evaluated by full-wave simulation (e.g. COMSOL or MEEP eigenmode solver, post-processing the tensor contraction in the volume of the meta-atoms). The ~ 0.3 figure is plausible for a properly engineered qBIC pair but is by no means generic – random mode pairings typically give η_{geom} of order 10^{-3} or less, especially when the SH-mode parity does not match the symmetric tensor contraction $\mathbf{f}_1\mathbf{f}_1$ at unit-cell level.

6. **Avoid the "single n" simplification.** Use $\widetilde{\varepsilon}(\mathbf{r}, \omega_1)$ and $\widetilde{\varepsilon}(\mathbf{r}, 2\omega_1)$ separately in the energy normalisation (8). The dispersion of $\chi^{(2)}$ between ω_1 and $2\omega_1$ is a few percent in $LiNbO_3$ but should be carried explicitly when comparing with experiment.

7. **Disorder and inhomogeneous broadening.** The Q_2 quoted throughout is the *element* quality factor. In an array of N meta-atoms with fabrication scatter σ_{ω_2} on the individual SH-resonance frequencies, the collective response is inhomogeneously broadened with an effective array linewidth $\gamma_2^{(\text{array})} \simeq \sqrt{\gamma_2^2 + \sigma_{\omega_2}^2}$, and the cascaded enhancement is set by $Q_2^{(\text{array})}$, not $Q_2^{(\text{element})}$. For $Q_2^{(\text{element})} \sim 10^4$ at $\omega_2/2\pi \sim 400$ THz, preserving the cascade requires $\sigma_{\omega_2}/\omega_2 \lesssim 10^{-4}$, i.e. sub-nanometre control of meta-atom dimensions. Mitigations include post-fabrication trimming (electron-beam-induced defects, photoresist trimming, thermal annealing), spatially averaged readout that recovers only the homogeneous response, or operating the SH cavity in an electro-optic feedback loop that pulls each element onto resonance.

14 Summary of the principal result

$$\boxed{\chi^{(3)}_{\text{casc}}(\omega_1) = -\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2\,\Delta\omega}\,\frac{\left|\int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\, d^3r\right|^2}{\int |\mathbf{f}_1(\mathbf{r})|^4\, d^3r} \;\simeq\; -\frac{2\,\omega_1}{3\,n_2^2\,\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\text{geom}}\,,} \tag{70}$$

valid for $\gamma_2/2 \ll |\Delta\omega| \ll \omega_1$ and $|\beta\, a_1| \ll |\Delta\omega|$, with β, α_3 as defined in Section 5 and N_2 the projection norm (10) ($= \frac{1}{2}$ in our normalisation). The first form is exact within single-mode theory and holds in any mode normalisation; the second defines the geometric overlap $\eta_{\text{geom}} \leq 1$ – only n_2 (the SH-mode effective index) survives, all n_1 powers cancelling between numerator and denominator – and reduces to the exact bulk bound-wave cascade $-2|\chi^{(2)}|^2/[3(n_2^2 - n_1^2)]$ for uniform phase-matched modes ($\eta_{\text{geom}} = 1$).

Compared with the original derivation, this version:

- fixes the overall prefactor (2/3, the DFG-to-SPM degeneracy ratio) and verifies it against the uniform-medium limit; an earlier version of the present note carried a factor-4 error here (Section 5);
- uses the full vector mode profile with proper energy normalisation (Brillouin) accounting for dispersion, rather than a scalar $f(\mathbf{r})$ with a single n;
- carries the tensorial $\chi^{(2)}_{ijk}$ (and $\chi^{(3)}_{ijkl}$) and gives an unambiguous overlap integral;
- includes the factor 2 in the DFG polarisation and the factor 3 in SPM, ensuring Manley–Rowe is satisfied and the SHG/DFG couplings carry the same coupling constant β;
- carries ε_0 correctly through the polarisation–field relation, so that the coupled-mode coefficients $\beta \propto \varepsilon_0\omega_1$ and $\alpha_3 \propto 3\varepsilon_0\omega_1$ are dimensionally clean; ε_0 then cancels in the final $\chi^{(3)}_{\text{casc}}$, as it must for a material parameter;
- makes the $1/n_2^2$ dependence explicit (the n_1 powers cancel between $|\int \chi^{(2)} f_2^* f_1^2|^2$ and $\int |\mathbf{f}_1|^4$, leaving only the SH-mode normalisation), correcting the $1/n^3$ scaling of the original note;
- includes both the cascaded Kerr *and* the cascaded two-photon-loss term, essential for the saturation behaviour;
- exposes the bandwidth–enhancement sum rule, the correct framing for comparison with bulk (or any non-resonant) Kerr-equivalent processes;
- adds the symmetry / phase-matching constraints that govern η in a metasurface platform.

Whether the resulting "fast Kerr from slow SHG" claim survives depends on the application: for narrowband, CW-pumped nonlinear photonics (bistability, frequency-comb interactions, low-threshold parametric oscillation) the metasurface route lowers the CW threshold power by orders of magnitude relative to bulk cascading, although bulk PPLN retains a $\sim$10$\times$ advantage in the integrated figure $|\chi^{(3)}| \cdot \Delta f$; for ultrafast applications the sum rule precludes any improvement over bulk.

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